

REQUEST FOR QUOTATION (THIS IS NOT AN ORDER)			THIS RFQ ☒ IS ☐ IS NOT A SMALL BUSINESS SET-ASIDE			PAGE 1 OF 58 PAGES	
1. REQUEST NUMBER 140F1S26Q0041		2. DATE ISSUED 04/27/2026		3. REQUISITION/PURCHASE REQUEST NUMBER 0044038212		4. CERT. FOR NAT. DEF. UNDER BDSA REG. 2 AND/OR DMS REG. 1	
5a. ISSUED BY FWS IT Services FWS, IT Services 5275 Leesburg Pike Falls Church VA 22041						6. DELIVER BY (Date)	
5b. FOR INFORMATION CALL (NO COLLECT CALLS)						7. DELIVERY ☒ FOB DESTINATION ☐ OTHER (See Schedule)	
NAME Melissa Niemi			TELEPHONE NUMBER AREA CODE 612 NUMBER 713-5216			9. DESTINATION	
8. TO:						a. NAME OF CONSIGNEE FWS SOUTHWESTERN NATIVE ARC	
a. NAME			b. COMPANY			b. STREET ADDRESS 7116 HATCHERY ROAD	
c. STREET ADDRESS						c. CITY DEXTER	
d. CITY			e. STATE		f. ZIP CODE		d. STATE NM e. ZIP CODE 88230-9608
10. PLEASE FURNISH QUOTATIONS TO THE ISSUING OFFICE IN BLOCK 5a ON OR BEFORE CLOSE OF BUSINESS (Date) 05/11/2026 1700 ED			IMPORTANT: This is a request for information and quotations furnished are not offers. If you are unable to quote, please so indicate on this form and return it to the address in Block 5a. This request does not commit the Government to pay any costs incurred in the preparation of the submission of this quotation or to contract for supplies or service. Supplies are of domestic origin unless otherwise indicated by quoter. Any representations and/or certifications attached to this Request for Quotation must be completed by the quoter.				
11. SCHEDULE (Include applicable Federal, State and local taxes)							
ITEM NUMBER (a)	SUPPLIES/SERVICES (b)			QUANTITY (c)	UNIT (d)	UNIT PRICE (e)	AMOUNT (f)
	The U.S. Fish and Wildlife Service (USFWS) Southwestern Native Aquatic Resources and Recovery Center in Dexter, NM, has a need to re-wire 5 buildings on site that have old and broken internal building wiring on CAT 3 cabling and upgrade everything to current construction standards of CAT 6a., under RFQ# 140F1S26Q0041. Please see the attached Statement of Work (SOW) for additional details. Period of Performance: 05/18/2026 to 11/17/2026 Continued...						
12. DISCOUNT FOR PROMPT PAYMENT				a. 10 CALENDAR DAYS (%)	b. 20 CALENDAR DAYS (%)	c. 30 CALENDAR DAYS (%)	d. CALENDAR DAYS NUMBER PERCENTAGE
NOTE: Additional provisions and representations ☐ are ☐ are not attached.							
13. NAME AND ADDRESS OF QUOTER				14. SIGNATURE OF PERSON AUTHORIZED TO SIGN QUOTATION		15. DATE OF QUOTATION	
a. NAME OF QUOTER				16. SIGNER		b. TELEPHONE	
b. STREET ADDRESS						AREA CODE	
c. COUNTY				a. NAME (Type or print)		NUMBER	
d. CITY				e. STATE f. ZIP CODE		c. TITLE (Type or print)	

NAME OF OFFEROR OR CONTRACTOR

ITEM NO. (A)	SUPPLIES/SERVICES (B)	QUANTITY (C)	UNIT (D)	UNIT PRICE (E)	AMOUNT (F)
00010	Remove Existing and Install New Cat 6 Cable				

52.252-2 Clauses Incorporated by Reference. (FEB 1998)

This contract incorporates one or more clauses by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. Also, the full text of a clause may be accessed electronically at this/these address(es):

For FAR Clauses: <http://acquisition.gov/far/index.html>

1- Clauses Incorporated by Reference

52.203-17 Contractor Employee Whistleblower Rights and Requirement To Inform Employees of Whistleblower Rights. (Nov 2023)

52.204-13 System for Award Management Maintenance (Oct 2018)

52.204-18 Commercial and Government Entity Code Maintenance (Aug 2020)

52.204-19 Incorporation by Reference of Representations and Certifications. (Dec 2014)

52.212-4 Contract Terms and Conditions-Commercial Items. (Nov 2023)

52.242-15 Stop-Work Order (Aug 1989)

52.227-14 Rights in Data-General (May 2014)

2- Full Text Clauses

52.212-5 Contract Terms and Conditions Required To Implement Statutes or Executive Orders—Commercial Products and Commercial Services.

As prescribed in [12.301\(b\)\(4\)](#), insert the following clause:

Contract Terms and Conditions Required To Implement Statutes or Executive Orders—Commercial Products and Commercial Services (Mar 2026)

(a) The Contractor shall comply with the following Federal Acquisition Regulation (FAR) clauses, which are incorporated in this contract by reference, to implement provisions of law or Executive orders applicable to acquisitions of commercial products and commercial services:

- (1) [52.203-19](#), Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements (Jan 2017) (section 743 of Division E, Title VII, of the Consolidated and Further Continuing Appropriations Act, 2015 (Pub. L. 113-235) and its successor provisions in subsequent appropriations acts (and as extended in continuing resolutions)).
- (2) [52.204-23](#), Prohibition on Contracting for Hardware, Software, and Services Developed or Provided by Kaspersky Lab Covered Entities (Dec 2023) (Section 1634 of Pub. L. 115-91).
- (3) [52.204-25](#), Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment. (Nov 2021) (Section 889(a)(1)(A) of Pub. L. 115-232).
- (4) [52.209-10](#), Prohibition on Contracting with Inverted Domestic Corporations (Nov 2015).
- (5) [52.232-40](#), Providing Accelerated Payments to Small Business Subcontractors (Mar 2023) ([31 U.S.C. 3903](#) and [10 U.S.C. 3801](#)).
- (6) [52.233-3](#), Protest After Award (Aug 1996) (31 U.S.C. 3553).

(7) [52.233-4](#), Applicable Law for Breach of Contract Claim (Oct 2004) (Public Laws 108-77 and 108-78 (19 U.S.C. 3805 note)).

(b) The Contractor shall comply with the FAR clauses in this paragraph (b) that the Contracting Officer has indicated as being incorporated in this contract by reference to implement provisions of law or Executive orders applicable to acquisitions of commercial products and commercial services:

[Contracting Officer check as appropriate.]

___ (1) [52.203-6](#), Restrictions on Subcontractor Sales to the Government (Jun 2020), with *Alternate I* (Nov 2021) (41 U.S.C. 4704 and [10 U.S.C. 4655](#)).

_X (2) [52.203-13](#), Contractor Code of Business Ethics and Conduct (Nov 2021) (41 U.S.C. 3509)).

___ (3) [52.203-15](#), Whistleblower Protections under the American Recovery and Reinvestment Act of 2009 (Jun 2010) (Section 1553 of Pub. L. 111-5). (Applies to contracts funded by the American Recovery and Reinvestment Act of 2009.)

___ (4) [52.203-17](#), Contractor Employee Whistleblower Rights (Nov 2023) ([41 U.S.C. 4712](#)); this clause does not apply to contracts of DoD, NASA, the Coast Guard, or applicable elements of the intelligence community—see FAR [3.900](#)(a).

___ (5) [52.204-10](#), Reporting Executive Compensation and First-Tier Subcontract Awards (Jun 2020) (Pub. L. 109-282) ([31 U.S.C. 6101 note](#)).

___ (6) [Reserved].

_X (7) [52.204-14](#), Service Contract Reporting Requirements (Oct 2016) (Pub. L. 111-117, section 743 of Div. C).

___ (8) [52.204-15](#), Service Contract Reporting Requirements for Indefinite-Delivery Contracts (Oct 2016) (Pub. L. 111-117, section 743 of Div. C).

_X (9) [52.204-27](#), Prohibition on a ByteDance Covered Application (Jun 2023) (Section 102 of Division R of Pub. L. 117-328).

___ (10) [52.204-28](#), Federal Acquisition Supply Chain Security Act Orders—Federal Supply Schedules, Governmentwide Acquisition Contracts, and Multi-Agency Contracts. (Dec 2023) ([Pub. L. 115-390](#), title II).

_X (11) (i) [52.204-30](#), Federal Acquisition Supply Chain Security Act Orders—Prohibition. (Dec 2023) ([Pub. L. 115-390](#), title II).

___ (ii) Alternate I (Dec 2023) of [52.204-30](#).

_X (12) [52.209-6](#), Protecting the Government's Interest When Subcontracting With Contractors Debarred, Suspended, Proposed for Debarment, or Voluntarily Excluded. (Jan 2025) ([31 U.S.C. 6101 note](#)).

___ (13) [52.209-9](#), Updates of Publicly Available Information Regarding Responsibility Matters (Oct 2018) ([41 U.S.C. 2313](#)).

___ (14) [Reserved].

___ (15) [52.219-3](#), Notice of HUBZone Set-Aside or Sole-Source Award (Oct 2022) ([15 U.S.C. 657a](#)).

___ (16) [52.219-4](#), Notice of Price Evaluation Preference for HUBZone Small Business Concerns (Oct 2022) (if the offeror elects to waive the preference, it shall so indicate in its offer) (15 U.S.C. 657a).

___ (17) [Reserved]

___ (18) (i) [52.219-6](#), Notice of Total Small Business Set-Aside (Nov 2020) (15 U.S.C. 644).

___ (ii) Alternate I (Mar 2020) of [52.219-6](#).

___ (19) (i) [52.219-7](#), Notice of Partial Small Business Set-Aside (Nov 2020) (15 U.S.C. 644).

___ (ii) Alternate I (Mar 2020) of [52.219-7](#).

___ (20) [52.219-8](#), Utilization of Small Business Concerns (Jan 2025)([15 U.S.C. 637](#)(d)(2) and (3)).

___ (21) (i) [52.219-9](#), Small Business Subcontracting Plan (Jan 2025) ([15 U.S.C. 637](#)(d)(4)).

___ (ii) Alternate I (Nov 2016) of [52.219-9](#).

___ (iii) Alternate II (Nov 2016) of [52.219-9](#).

___ (iv) Alternate III (Jun 2020) of [52.219-9](#).

___ (v) Alternate IV (Jan 2025) of [52.219-9](#).

___ (22) (i) [52.219-13](#), Notice of Set-Aside of Orders (Mar 2020) (15 U.S.C. 644(r)).

___ (ii) Alternate I (Mar 2020) of [52.219-13](#).

- ___ (23) [52.219-14](#), Limitations on Subcontracting (Oct 2022) (15 U.S.C. 657s).
- ___ (24) [52.219-16](#), Liquidated Damages—Subcontracting Plan (Sep 2021) (15 U.S.C. 637(d)(4)(F)(i)).
- ___ (25) [52.219-27](#), Notice of Set-Aside for, or Sole-Source Award to, Service-Disabled Veteran-Owned Small Business (SDVOSB) Concerns Eligible Under the SDVOSB Program (Feb 2024) (15 U.S.C. 657f).
- ___ (26) (i) [52.219-28](#), Postaward Small Business Program Rerepresentation (Jan 2025) (15 U.S.C. 632(a)(2)).
- ___ (ii) Alternate I (Mar 2020) of [52.219-28](#).
- ___ (27) [52.219-29](#), Notice of Set-Aside for, or Sole-Source Award to, Economically Disadvantaged Women-Owned Small Business Concerns (Oct 2022) (15 U.S.C. 637(m)).
- ___ (28) [52.219-30](#), Notice of Set-Aside for, or Sole-Source Award to, Women-Owned Small Business Concerns Eligible Under the Women-Owned Small Business Program (Oct 2022) (15 U.S.C. 637(m)).
- ___ (29) [52.219-32](#), Orders Issued Directly Under Small Business Reserves (Mar 2020) ([15 U.S.C. 644\(r\)](#)).
- ___ (30) [52.219-33](#), Nonmanufacturer Rule (Sep 2021) ([15 U.S.C. 637](#)(a)(17)).
- _X (31) [52.222-3](#), Convict Labor (Jun 2003) (E.O. 11755).
- ___ (32) [52.222-19](#), Child Labor—Cooperation with Authorities and Remedies (Mar 2026)([E.O. 13126](#)).
- _X (33) [52.222-21](#), Prohibition of Segregated Facilities (Apr 2015).
- _X (34) (i) [52.222-26](#), Equal Opportunity (Sep 2016) (E.O. 11246).
- ___ (ii) Alternate I (Feb 1999) of [52.222-26](#).
- _X (35) (i) [52.222-35](#), Equal Opportunity for Veterans (Jun 2020) ([38 U.S.C. 4212](#)).
- ___ (ii) Alternate I (Jul 2014) of [52.222-35](#).
- _X (36) (i) [52.222-36](#), Equal Opportunity for Workers with Disabilities (Jun 2020) ([29 U.S.C. 793](#)).
- ___ (ii) Alternate I (Jul 2014) of [52.222-36](#).
- ___ (37) [52.222-37](#), Employment Reports on Veterans (Jun 2020) ([38 U.S.C. 4212](#)).
- ___ (38) [52.222-40](#), Notification of Employee Rights Under the National Labor Relations Act (Dec 2010) (E.O. 13496).
- _X (39) (i) [52.222-50](#), Combating Trafficking in Persons (Oct 2025) (22 U.S.C. chapter 78 and E.O. 13627).
- ___ (ii) Alternate I (Mar 2015) of [52.222-50](#) (22 U.S.C. chapter 78 and E.O. 13627).
- ___ (40) [52.222-54](#), Employment Eligibility Verification (Jan 2025) ([Executive Order 12989](#)). (Not applicable to the acquisition of commercially available off-the-shelf items or certain other types of commercial products or commercial services as prescribed in FAR [22.1803](#).)
- ___ (41) (i) [52.223-9](#), Estimate of Percentage of Recovered Material Content for EPA–Designated Items (May 2008) (42 U.S.C. 6962(c)(3)(A)(ii)). (Not applicable to the acquisition of commercially available off-the-shelf items.)
- ___ (ii) Alternate I (May 2008) of [52.223-9](#) (42 U.S.C. 6962(i)(2)(C)). (Not applicable to the acquisition of commercially available off-the-shelf items.)
- ___ (42) [52.223-11](#), Ozone-Depleting Substances and High Global Warming Potential Hydrofluorocarbons (May 2024) ([42 U.S.C. 7671](#), *et seq.*).
- ___ (43) [52.223-12](#), Maintenance, Service, Repair, or Disposal of Refrigeration Equipment and Air Conditioners (May 2024) ([42 U.S.C. 7671](#), *et seq.*).
- ___ (44) [52.223-20](#), Aerosols (May 2024) ([42 U.S.C. 7671](#), *et seq.*).
- ___ (45) [52.223-21](#), Foams (May 2024) ([42 U.S.C. 7671](#), *et seq.*).
- ___ (46) [52.223-23](#), Sustainable Products and Services (May 2024) ([E.O. 14057](#), [7 U.S.C. 8102](#), [42 U.S.C. 6962](#), [42 U.S.C. 8259b](#), and [42 U.S.C. 7671i](#)).
- ___ (47) (i) [52.224-3](#) Privacy Training (Jan 2017) ([5 U.S.C. 552](#) a).
- ___ (ii) Alternate I (Jan 2017) of [52.224-3](#).
- ___ (48) (i) [52.225-1](#), Buy American-Supplies (Oct 2022) (41 U.S.C. chapter 83).
- ___ (ii) Alternate I (Oct 2022) of [52.225-1](#).
- ___ (49) (i) [52.225-3](#), Buy American-Free Trade Agreements-Israeli Trade Act (NOV 2023) ([19 U.S.C. 3301 note](#), [19 U.S.C. 2112 note](#), [19 U.S.C. 3805 note](#), [19 U.S.C. 4001 note](#), [19 U.S.C. chapter](#)

[29](#) (sections 4501-4732), Public Law 103-182, 108-77, 108-78, 108-286, 108-302, 109-53, 109-169, 109-283, 110-138, 112-41, 112-42, and 112-43.

___ (ii) Alternate I [Reserved].

___ (iii) Alternate II (Jan 2025) of [52.225-3](#).

___ (iv) Alternate III (Feb 2024) of [52.225-3](#).

___ (v) Alternate IV (Oct 2022) of [52.225-3](#).

___ (50) [52.225-5](#), Trade Agreements (NOV 2023) ([19 U.S.C. 2501](#), *et seq.*, [19 U.S.C. 3301](#) note).

___ (51) [52.225-13](#), Restrictions on Certain Foreign Purchases (Feb 2021) (E.O.'s, proclamations, and statutes administered by the Office of Foreign Assets Control of the Department of the Treasury).

___ (52) [52.225-26](#), Contractors Performing Private Security Functions Outside the United States (Oct 2016) (Section 862, as amended, of the National Defense Authorization Act for Fiscal Year 2008; 10 U.S.C. Subtitle A, Part V, Subpart G Note).

___ (53) [52.226-4](#), Notice of Disaster or Emergency Area Set-Aside (Nov 2007) (42 U.S.C. 5150).

___ (54) [52.226-5](#), Restrictions on Subcontracting Outside Disaster or Emergency Area (Nov 2007) (42 U.S.C. 5150).

_X (55) [52.226-8](#), Encouraging Contractor Policies to Ban Text Messaging While Driving (May 2024) ([E.O. 13513](#)).

___ (56) [52.229-12](#), Tax on Certain Foreign Procurements (Feb 2021).

___ (57) [52.232-29](#), Terms for Financing of Purchases of Commercial Products and Commercial Services (Nov 2021) (41 U.S.C. 4505, [10 U.S.C. 3805](#)).

___ (58) [52.232-30](#), Installment Payments for Commercial Products and Commercial Services (Nov 2021) (41 U.S.C. 4505, [10 U.S.C. 3805](#)).

_X (59) [52.232-33](#), Payment by Electronic Funds Transfer-System for Award Management (Oct 2018) ([31 U.S.C. 3332](#)).

___ (60) [52.232-34](#), Payment by Electronic Funds Transfer-Other than System for Award Management (Jul 2013) (31 U.S.C. 3332).

___ (61) [52.232-36](#), Payment by Third Party (May 2014) (31 U.S.C. 3332).

___ (62) [52.239-1](#), Privacy or Security Safeguards (Aug 1996) ([5 U.S.C. 552a](#)).

___ (63) [52.240-1](#), Prohibition on Unmanned Aircraft Systems Manufactured or Assembled by American Security Drone Act-Covered Foreign Entities (Nov 2024) (Sections 1821-1826, Pub. L. 118-31, [41 U.S.C. 3901](#) note prec.).

___ (64) [52.242-5](#), Payments to Small Business Subcontractors (Jan 2017) (15 U.S.C. 637(d)(13)).

___ (65) (i) [52.247-64](#), Preference for Privately Owned U.S.-Flag Commercial Vessels (Nov 2021) ([46 U.S.C. 55305](#) and 10 U.S.C. 2631).

___ (ii) Alternate I (Apr 2003) of [52.247-64](#).

___ (iii) Alternate II (Nov 2021) of [52.247-64](#).

(c) The Contractor shall comply with the FAR clauses in this paragraph (c), applicable to commercial services, that the Contracting Officer has indicated as being incorporated in this contract by reference to implement provisions of law or Executive orders applicable to acquisitions of commercial products and commercial services:

[Contracting Officer check as appropriate.]

___ (1) [52.222-41](#), Service Contract Labor Standards (Aug 2018) (41 U.S.C. chapter 67).

___ (2) [52.222-42](#), Statement of Equivalent Rates for Federal Hires (May 2014) (29 U.S.C. 206 and 41 U.S.C. chapter 67).

___ (3) [52.222-43](#), Fair Labor Standards Act and Service Contract Labor Standards-Price Adjustment (Multiple Year and Option Contracts) (Aug 2018) (29 U.S.C. 206 and 41 U.S.C. chapter 67).

___ (4) [52.222-44](#), Fair Labor Standards Act and Service Contract Labor Standards-Price Adjustment (May 2014) ([29 U.S.C. 206](#) and 41 U.S.C. chapter 67).

___ (5) [52.222-51](#), Exemption from Application of the Service Contract Labor Standards to Contracts for Maintenance, Calibration, or Repair of Certain Equipment-Requirements (May 2014) (41 U.S.C. chapter 67).

___ (6) [52.222-53](#), Exemption from Application of the Service Contract Labor Standards to Contracts for Certain Services-Requirements (May 2014) (41 U.S.C. chapter 67).

___ (7) [52.222-55](#), Minimum Wages for Contractor Workers Under Executive Order 14026 (Jan 2022).

___ (8) [52.222-62](#), Paid Sick Leave Under Executive Order 13706 (Jan 2022) (E.O. 13706).

___ (9) [52.226-6](#), Promoting Excess Food Donation to Nonprofit Organizations (Jun 2020) (42 U.S.C. 1792).

___ (10) [52.247-69](#), Reporting Requirement for U.S.-Flag Air Carriers Regarding Training to Prevent Human Trafficking (Jan 2025) ([49 U.S.C. 40118\(g\)](#)).

(d) *Comptroller General Examination of Record*. The Contractor shall comply with the provisions of this paragraph (d) if this contract was awarded using other than sealed bid, is in excess of the simplified acquisition threshold, as defined in FAR [2.101](#), on the date of award of this contract, and does not contain the clause at [52.215-2](#), Audit and Records-Negotiation.

(1) The Comptroller General of the United States, or an authorized representative of the Comptroller General, shall have access to and right to examine any of the Contractor's directly pertinent records involving transactions related to this contract.

(2) The Contractor shall make available at its offices at all reasonable times the records, materials, and other evidence for examination, audit, or reproduction, until 3 years after final payment under this contract or for any shorter period specified in FAR subpart [4.7](#), Contractor Records Retention, of the other clauses of this contract. If this contract is completely or partially terminated, the records relating to the work terminated shall be made available for 3 years after any resulting final termination settlement. Records relating to appeals under the disputes clause or to litigation or the settlement of claims arising under or relating to this contract shall be made available until such appeals, litigation, or claims are finally resolved.

(3) As used in this clause, records include books, documents, accounting procedures and practices, and other data, regardless of type and regardless of form. This does not require the Contractor to create or maintain any record that the Contractor does not maintain in the ordinary course of business or pursuant to a provision of law.

(e)

(1) Notwithstanding the requirements of the clauses in paragraphs (a), (b), (c), and (d) of this clause, the Contractor is not required to flow down any FAR clause, other than those in this paragraph (e)(1), in a subcontract for commercial products or commercial services. Unless otherwise indicated below, the extent of the flow down shall be as required by the clause-

(i) [52.203-13](#), Contractor Code of Business Ethics and Conduct (Nov 2021) (41 U.S.C. 3509).

(ii) [52.203-17](#), Contractor Employee Whistleblower Rights (Nov 2023) ([41 U.S.C. 4712](#)).

(iii) [52.203-19](#), Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements (Jan 2017) (section 743 of Division E, Title VII, of the Consolidated and Further Continuing Appropriations Act, 2015 (Pub. L. 113-235) and its successor provisions in subsequent appropriations acts (and as extended in continuing resolutions)).

(iv) [52.204-23](#), Prohibition on Contracting for Hardware, Software, and Services Developed or Provided by Kaspersky Lab Covered Entities (Dec 2023) (Section 1634 of Pub. L. 115-91).

(v) [52.204-25](#), Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment. (Nov 2021) (Section 889(a)(1)(A) of Pub. L. 115-232).

(vi) [52.204-27](#), Prohibition on a ByteDance Covered Application (Jun 2023) (Section 102 of Division R of Pub. L. 117-328).

(vii)

(A) 52.204-30, Federal Acquisition Supply Chain Security Act Orders—Prohibition. (Dec 2023) ([Pub. L. 115-390](#), title II).

(B) Alternate I (Dec 2023) of 52.204-30.

(viii) [52.219-8](#), Utilization of Small Business Concerns (Jan 2025) ([15 U.S.C. 637\(d\)\(2\)](#) and (3)), in all subcontracts that offer further subcontracting opportunities. If the subcontract (except subcontracts to small business concerns) exceeds the applicable threshold specified in FAR [19.702\(a\)](#) on the date of

subcontract award, the subcontractor must include [52.219-8](#) in lower tier subcontracts that offer subcontracting opportunities.

- (ix) [52.222-21](#), Prohibition of Segregated Facilities (Apr 2015).
- (x) [52.222-26](#), Equal Opportunity (Sep 2016) (E.O. 11246).
- (xi) [52.222-35](#), Equal Opportunity for Veterans (Jun 2020) (38 U.S.C. 4212).
- (xii) [52.222-36](#), Equal Opportunity for Workers with Disabilities (Jun 2020) (29 U.S.C. 793).
- (xiii) [52.222-37](#), Employment Reports on Veterans (Jun 2020) (38 U.S.C. 4212).
- (xiv) [52.222-40](#), Notification of Employee Rights Under the National Labor Relations Act (Dec 2010) (E.O. 13496). Flow down required in accordance with paragraph (f) of FAR clause [52.222-40](#).
- (xv) [52.222-41](#), Service Contract Labor Standards (Aug 2018) ([41 U.S.C. chapter 67](#)).
- (xvi)
 - (A) [52.222-50](#), Combating Trafficking in Persons (Oct 2025) (22 U.S.C. chapter 78 and E.O. 13627).
 - (B) Alternate I (Mar 2015) of [52.222-50](#) (22 U.S.C. chapter 78 and E.O. 13627).
- (xvii) [52.222-51](#), Exemption from Application of the Service Contract Labor Standards to Contracts for Maintenance, Calibration, or Repair of Certain Equipment-Requirements (May 2014) (41 U.S.C. chapter 67).
- (xviii) [52.222-53](#), Exemption from Application of the Service Contract Labor Standards to Contracts for Certain Services-Requirements (May 2014) (41 U.S.C. chapter 67).
- (xix) [52.222-54](#), Employment Eligibility Verification (Jan 2025) ([E.O. 12989](#)).
- (xx) [52.222-55](#), Minimum Wages for Contractor Workers Under Executive Order 14026 (Jan 2022).
- (xxi) [52.222-62](#), Paid Sick Leave Under Executive Order 13706 (Jan 2022) (E.O. 13706).
- (xxii)
 - (A) [52.224-3](#), Privacy Training (Jan 2017) ([5 U.S.C. 552a](#)).
 - (B) Alternate I (Jan 2017) of [52.224-3](#).
- (xxiii) [52.225-26](#), Contractors Performing Private Security Functions Outside the United States (Oct 2016) (Section 862, as amended, of the National Defense Authorization Act for Fiscal Year 2008; 10 U.S.C. Subtitle A, Part V, Subpart G Note).
- (xxiv) [52.226-6](#), Promoting Excess Food Donation to Nonprofit Organizations (Jun 2020) (42 U.S.C. 1792). Flow down required in accordance with paragraph (e) of FAR clause [52.226-6](#).
- (xxv) [52.232-40](#), Providing Accelerated Payments to Small Business Subcontractors (Mar 2023) ([31 U.S.C. 3903](#) and [10 U.S.C. 3801](#)). Flow down required in accordance with paragraph (c) of [52.232-40](#).
- (xxvi) [52.240-1](#), Prohibition on Unmanned Aircraft Systems Manufactured or Assembled by American Security Drone Act-Covered Foreign Entities (Nov 2024) (Sections 1821-1826, Pub. L. 118-31, [41 U.S.C. 3901](#) note prec.).
- (xxvii) [52.247-64](#), Preference for Privately Owned U.S.-Flag Commercial Vessels (Nov 2021) ([46 U.S.C. 55305](#) and 10 U.S.C. 2631). Flow down required in accordance with paragraph (d) of FAR clause [52.247-64](#).
- (2) While not required, the Contractor may include in its subcontracts for commercial products and commercial services a minimal number of additional clauses necessary to satisfy its contractual obligations.
- (End of clause)

52.217-6 Option for Increased Quantity.

As prescribed in [17.208](#)(d), insert a clause substantially the same as the following:

Option for Increased Quantity (Mar 1989)

The Government may increase the quantity of supplies called for in the Schedule at the unit price specified. The Contracting Officer may exercise the option by written notice to the Contractor within 60-

90 Days of the needed increase. Delivery of the added items shall continue at the same rate as the like items called for under the contract, unless the parties otherwise agree.
(End of clause)

52.217-7 Option for Increased Quantity-Separately Priced Line Item.

As prescribed in [17.208](#)(e), insert a clause substantially the same as the following:

Option for Increased Quantity-Separately Priced Line Item (Mar 1989)

The Government may require the delivery of the numbered line item, identified in the Schedule as an option item, in the quantity and at the price stated in the Schedule. The Contracting Officer may exercise the option by written notice to the Contractor within 60-90 Days of the needed increase. Delivery of added items shall continue at the same rate that like items are called for under the contract, unless the parties otherwise agree.

(End of clause)

52.217-8 Option to Extend Services.

As prescribed in [17.208](#)(f), insert a clause substantially the same as the following:

OPTION TO EXTEND SERVICES (NOV 1999)

The Government may require continued performance of any services within the limits and at the rates specified in the contract. These rates may be adjusted only as a result of revisions to prevailing labor rates provided by the Secretary of Labor. The option provision may be exercised more than once, but the total extension of performance hereunder shall not exceed 6 months. The Contracting Officer may exercise the option by written notice to the Contractor within *30 days*.

(End of clause)

52.204-25 Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment.

As prescribed in [4.2105](#)(b), insert the following clause:

PROHIBITION ON CONTRACTING FOR CERTAIN TELECOMMUNICATIONS AND VIDEO SURVEILLANCE SERVICES OR EQUIPMENT (NOV 2021)

(a) *Definitions.* As used in this clause—

Backhaul means intermediate links between the core network, or backbone network, and the small subnetworks at the edge of the network (*e.g.*, connecting cell phones/towers to the core telephone network). Backhaul can be wireless (*e.g.*, microwave) or wired (*e.g.*, fiber optic, coaxial cable, Ethernet).

Covered foreign country means The People's Republic of China.

Covered telecommunications equipment or services means—

- (1) Telecommunications equipment produced by Huawei Technologies Company or ZTE Corporation (or any subsidiary or affiliate of such entities);
- (2) For the purpose of public safety, security of Government facilities, physical security surveillance of critical infrastructure, and other national security purposes, video surveillance and telecommunications equipment produced by Hytera Communications Corporation, Hangzhou Hikvision Digital Technology Company, or Dahua Technology Company (or any subsidiary or affiliate of such entities);
- (3) Telecommunications or video surveillance services provided by such entities or using such equipment; or
- (4) Telecommunications or video surveillance equipment or services produced or provided by an entity that the Secretary of Defense, in consultation with the Director of National Intelligence or the Director of the Federal Bureau of Investigation, reasonably believes to be an entity owned or controlled by, or otherwise connected to, the government of a covered foreign country.

Critical technology means—

- (1) Defense articles or defense services included on the United States Munitions List set forth in the International Traffic in Arms Regulations under subchapter M of chapter I of title 22, Code of Federal Regulations;
- (2) Items included on the Commerce Control List set forth in Supplement No. 1 to part 774 of the Export Administration Regulations under subchapter C of chapter VII of title 15, Code of Federal Regulations, and controlled-
 - (i) Pursuant to multilateral regimes, including for reasons relating to national security, chemical and biological weapons proliferation, nuclear nonproliferation, or missile technology; or
 - (ii) For reasons relating to regional stability or surreptitious listening;
- (3) Specially designed and prepared nuclear equipment, parts and components, materials, software, and technology covered by part 810 of title 10, Code of Federal Regulations (relating to assistance to foreign atomic energy activities);
- (4) Nuclear facilities, equipment, and material covered by part 110 of title 10, Code of Federal Regulations (relating to export and import of nuclear equipment and material);
- (5) Select agents and toxins covered by part 331 of title 7, Code of Federal Regulations, part 121 of title 9 of such Code, or part 73 of title 42 of such Code; or
- (6) Emerging and foundational technologies controlled pursuant to section 1758 of the Export Control Reform Act of 2018 (50 U.S.C. 4817).

Interconnection arrangements means arrangements governing the physical connection of two or more networks to allow the use of another's network to hand off traffic where it is ultimately delivered (e.g.,

connection of a customer of telephone provider A to a customer of telephone company B) or sharing data and other information resources.

Reasonable inquiry means an inquiry designed to uncover any information in the entity's possession about the identity of the producer or provider of covered telecommunications equipment or services used by the entity that excludes the need to include an internal or third-party audit.

Roaming means cellular communications services (e.g., voice, video, data) received from a visited network when unable to connect to the facilities of the home network either because signal coverage is too weak or because traffic is too high.

Substantial or essential component means any component necessary for the proper function or performance of a piece of equipment, system, or service.

(b) Prohibition.

(1) Section 889(a)(1)(A) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (Pub. L. 115-232) prohibits the head of an executive agency on or after August 13, 2019, from procuring or obtaining, or extending or renewing a contract to procure or obtain, any equipment, system, or service that uses covered telecommunications equipment or services as a substantial or essential component of any system, or as critical technology as part of any system. The Contractor is prohibited from providing to the Government any equipment, system, or service that uses covered telecommunications equipment or services as a substantial or essential component of any system, or as critical technology as part of any system, unless an exception at paragraph (c) of this clause applies or the covered telecommunication equipment or services are covered by a waiver described in FAR [4.2104](#).

(2) Section 889(a)(1)(B) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (Pub. L. 115-232) prohibits the head of an executive agency on or after August 13, 2020, from entering into a contract, or extending or renewing a contract, with an entity that uses any equipment, system, or service that uses covered telecommunications equipment or services as a substantial or essential component of any system, or as critical technology as part of any system, unless an exception at paragraph (c) of this clause applies or the covered telecommunication equipment or services are covered by a waiver described in FAR [4.2104](#). This prohibition applies to the use of covered telecommunications equipment or services, regardless of whether that use is in performance of work under a Federal contract.

(c) Exceptions. This clause does not prohibit contractors from providing—

(1) A service that connects to the facilities of a third-party, such as backhaul, roaming, or interconnection arrangements; or

(2) Telecommunications equipment that cannot route or redirect user data traffic or permit visibility into any user data or packets that such equipment transmits or otherwise handles.

(d) Reporting requirement.

(1) In the event the Contractor identifies covered telecommunications equipment or services used as a substantial or essential component of any system, or as critical technology as part of any system, during contract performance, or the Contractor is notified of such by a subcontractor at any tier or by any other source, the Contractor shall report the information in paragraph (d)(2) of this clause to the Contracting

Officer, unless elsewhere in this contract are established procedures for reporting the information; in the case of the Department of Defense, the Contractor shall report to the website at <https://dibnet.dod.mil>. For indefinite delivery contracts, the Contractor shall report to the Contracting Officer for the indefinite delivery contract and the Contracting Officer(s) for any affected order or, in the case of the Department of Defense, identify both the indefinite delivery contract and any affected orders in the report provided at <https://dibnet.dod.mil>.

(2) The Contractor shall report the following information pursuant to paragraph (d)(1) of this clause

(i) Within one business day from the date of such identification or notification: the contract number; the order number(s), if applicable; supplier name; supplier unique entity identifier (if known); supplier Commercial and Government Entity (CAGE) code (if known); brand; model number (original equipment manufacturer number, manufacturer part number, or wholesaler number); item description; and any readily available information about mitigation actions undertaken or recommended.

(ii) Within 10 business days of submitting the information in paragraph (d)(2)(i) of this clause: any further available information about mitigation actions undertaken or recommended. In addition, the Contractor shall describe the efforts it undertook to prevent use or submission of covered telecommunications equipment or services, and any additional efforts that will be incorporated to prevent future use or submission of covered telecommunications equipment or services.

(e) *Subcontracts*. The Contractor shall insert the substance of this clause, including this paragraph (e) and excluding paragraph (b)(2), in all subcontracts and other contractual instruments, including subcontracts for the acquisition of commercial products or commercial services.

(End of clause)

DEPARTMENT OF THE INTERIOR ACQUISITION REGULATION (DIAR) CLAUSES INCORPORATED BY FULL TEXT

1452.201-70 Authorities and Delegations (Sep 2011)

(a) The Contracting Officer is the only individual authorized to enter into or terminate this contract, modify any term or condition of this contract, waive any requirement of this contract, or accept nonconforming work.

(b) The Contracting Officer will designate a Contracting Officer's Representative (COR) at time of award. The COR will be responsible for technical monitoring of the contractor's performance and deliveries. The COR will be appointed in writing, and a copy of the appointment will be furnished to the Contractor. Changes to this delegation will be made by written changes to the existing appointment or by issuance of a new appointment.

(c) The COR is not authorized to perform, formally or informally, any of the following actions:

(1) Promise, award, agree to award, or execute any contract, contract modification, or notice of intent that changes or may change this contract;

(2) Waive or agree to modification of the delivery schedule;

(3) Make any final decision on any contract matter subject to the Disputes Clause;

(4) Terminate, for any reason, the Contractor's right to proceed;

(5) Obligate in any way, the payment of money by the Government.

(d) The Contractor shall comply with the written or oral direction of the Contracting Officer or authorized representative(s) acting within the scope and authority of the appointment memorandum. The Contractor need not proceed with direction that it considers to have been issued without proper authority. The Contractor shall notify the Contracting Officer in writing, with as much detail as possible, when the COR has taken an action or has issued direction (written or oral) that the Contractor considers to exceed the COR's appointment, within 3 days of the occurrence. Unless otherwise provided in this contract, the Contractor assumes all costs, risks, liabilities, and consequences of performing any work it is directed to perform that falls within any of the categories defined in paragraph (c) prior to receipt of the Contracting Officer's response issued under paragraph (e) of this clause.

(e) The Contracting Officer shall respond in writing within 30 days to any notice made under paragraph (d) of this clause. A failure of the parties to agree upon the nature of a direction, or upon the contract action to be taken with respect thereto, shall be subject to the provisions of the Disputes clause of this contract.

(f) The Contractor shall provide copies of all correspondence to the Contracting Officer and the COR.

(g) Any action(s) taken by the Contractor, in response to any direction given by any person acting on behalf of the Government or any Government official other than the Contracting Officer or the COR acting within his or her appointment, shall be at the Contractor's risk.

(End of clause)

Electronic Invoicing and Payment Requirements - Invoice Processing Platform (IPP)

(February 2021)

Payment requests must be submitted electronically through the U. S. Department of the Treasury's Invoice Processing Platform System (IPP).

"Payment request" means any request for contract financing payment or invoice payment by the Contractor. To constitute a proper invoice, the payment request must comply with the requirements identified in the applicable Prompt Payment clause included in the contract, or the clause 52.212-4 Contract Terms and Conditions - Commercial Items included in commercial item contracts. The IPP website address is: <https://www.ipp.gov>.

Under this contract, the following documents are required to be submitted as an attachment to the IPP invoice [Contracting Officer to edit and include the documentation required under this contract]:

The Contractor must use the IPP website to register access and use IPP for submitting requests for payment. The Contractor Government Business Point of Contact (as listed in SAM) will receive enrollment instructions via email from the Federal Reserve Bank of St. Louis (FRBSTL) within 3 - 5 business days of the contract award date. Contractor assistance with enrollment can be obtained by

contacting the IPP Production Helpdesk via email IPPCustomerSupport@fiscal.treasury.gov or phone (866) 973-3131.

If the Contractor is unable to comply with the requirement to use IPP for submitting invoices for payment, the Contractor must submit a waiver request in writing to the Contracting Officer with its proposal or quotation.

(End of Local Clause)

INTERNET PROTOCOL version 6 (June 2012)

1. Any system hardware, software, firmware and/or networked component (voice, video or data) developed, procured, or acquired in support and/or performance of this contract shall be capable of transmitting, receiving, processing, forwarding and storing digital information across system boundaries utilizing system packets that are formatted in accordance with commercial standards of Internet Protocol (IP) version 6 (IPv6) as set forth in the USGv6 Profile ([NIST Special Publication 500267](#)) and corresponding declarations of conformance defined in the USGv6 Test Program. In addition, this system shall maintain interoperability with IPv4 systems and provide at least the same level of performance and reliability capabilities of IPv4 products.
2. Specifically, any new IP product or system developed, acquired, or produced must:
 - a. Interoperate with both IPv6 and IPv4 systems and products, and
 - b. Have available contractor/vendor IPv6 technical support for development and implementation and fielded product management.
3. As IPv6 evolves, the Contractor commits to upgrading or providing an appropriate migration path for each item developed, delivered or utilized at no additional cost to the Government. The Contractor shall retrofit all nonIPv6 capable equipment, as defined above, that is fielded under this contract with IPv6 capable equipment, at no additional cost to the Government.
4. The contractor shall provide technical support for both IPv4 and IPv6.
5. Any system or software must be able to operate on networks supporting IPv4, IPv6 or one that supports both.
6. Any product whose noncompliance is discovered and made known to the Contractor within one year after acceptance shall be upgraded, modified, or replaced to bring it into compliance at no additional cost to the Government.

(End of clause)

1452.233-2 -- Service of Protest Department of the Interior (Jul 1996) (Deviation)

- (a) Protests, as defined in section 33.101 of the Federal Acquisition Regulation, that are filed directly with an agency, and copies of any protests that are filed with the Government Accountability Office (GAO), shall be served on the Contracting Officer (addressed as follows) by obtaining written and dated acknowledgment of receipt from U.S. Fish & Wildlife, 4301 Fairfax Drive, Arlington, VA 22203.
- (b) The copy of any protest shall be received in the office designated above within one day of filing a protest with the GAO.
- (c) A copy of the protest served on the Contracting Officer shall be simultaneously furnished by the protester to the Department of the Interior Assistant Solicitor, Acquisitions and Intellectual Property, 1849 C Street, NW, Room 6511, Washington, DC 20240.

(End of Provision)

52.252-1 Solicitation Provisions Incorporated by Reference (Feb 1998)

This solicitation incorporates one or more solicitation provisions by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. The offeror is cautioned that the listed provisions may include blocks that must be completed by the offeror and submitted with its quotation or offer. In lieu of submitting the full text of those provisions, the offeror may identify the provision by paragraph identifier and provide the appropriate information with its quotation or offer. Also, the full text of a solicitation provision may be accessed electronically at this address: www.acquisition.gov.

Solicitations Incorporated by Reference

52.204-7 System for Award Management (Nov 2024)

52.204-16 Commercial and Government Entity Code Reporting (Aug 2020)

52.204-24 Representation Regarding Certain Telecommunications and Video Surveillance Services or Equipment. (Nov 2021)

52.204-26 Covered Telecommunications Equipment or Services-Representation (Oct 2020)

52.204-29 Federal Acquisition Supply Chain Security Act Orders-Representation and Disclosures (DEC 2023)

52.209-2 Prohibition on Contracting with Inverted Domestic Corporations-Representation (Nov 2015)

52.212-3 Offeror Representations and Certifications – Commercial Products and Commercial Services (OCT 2025)

52.225-25 Prohibition on Contracting with Entities Engaging in Sanctioned Activities Relating to Iran-Representation and Certifications (June 2020)



**Information Resources and Technology Management
(IRTM)**

**FWS AGENCY SPECIFIC REQUIREMENTS
FOR IT INFRASTRUCTURE**

March 2024 Version 2.0

STATEMENT OF WORK
)

1. General:

1.1 Southwestern Native Aquatic Resources and Recovery Center

Purpose/End Application:

To re-wire 5 buildings on site that have old and broken internal building wiring on CAT 3 cabling and upgrade everything to current construction standards of CAT 6a. The total number of ports is approximately 26 with the main (internet source) building having 14 ports. The rest of the buildings have the current wiring supplied via underground conduits: Fish Culture Building (3 ports), Fish Health Office Building (5 ports), Virology/Necropsy (2 ports), and Maintenance (2 ports). Test and verify from patch panel to wall jack that current standards are met and that network connectivity is achieved.

2. Place of Performance and Period of Performance

2.1 Infrastructure installation will take place on site at Southwestern Native Aquatic Resources and Recovery Center, Dexter, NM

2.2. Normal Business hours Monday thru Friday 6:30 a.m. - 4:00 p.m. Gate closes at 4:00 p.m.

2.3 Normal duty hours Monday thru Friday 6:30 a.m. - 4:00 p.m. In the event of emergencies, Contractor shall provide contact information.

2.4 Nonduty hours include weeknights, weekends, and Government holidays (New Year's Day, Martin Luther King JR's Birthday, Memorial Day, Independence Day, Juneteenth, Labor Day, Columbus Day, Veteran's Day, Thanksgiving Day, Christmas Day).

3. Scope

3.1 Contractor shall:

Refer to section 2.0 of the attached FWS Agency Specific Requirements for IT Infrastructure documentation.

Cabling Assessment & Planning:

Conduct a comprehensive site survey to identify current network layout, device locations, and areas requiring cabling replacement or addition. Develop a detailed network map outlining cable pathways, termination points, and equipment locations.

Cable Installation:

Remove old CAT 3 ethernet cables. Install new CAT 6a ethernet cables Per IRTM Standards. Run cables through walls, ceilings, and other designated pathways, utilizing conduits or cable trays as needed to protect and organize the cabling. Properly terminate and punch down cables at patch

panels, wall jacks, and other connection points, adhering IRTM documented construction wiring standards.

Testing & Certification:

Provide certification documentation of test results, including a cable map and labeling scheme with a LAN Room Cat6 Cable Test Report referenced on page 21 of the attached FWS Agency Specific Requirements for IT Infrastructure documentation.

Labeling & Documentation:

Clearly label both ends of each newly installed cable with a consistent and easily identifiable labeling scheme as described on page 20 of the attached FWS Agency Specific Requirements for IT Infrastructure documentation. Update existing network documentation to reflect the new cabling infrastructure, including a cable map, labeling scheme, and test results.

Cleanup:

Remove all debris and waste materials resulting from the installation process, leaving the work area clean and organized.

4. Deliverables and/or Delivery Schedule:

The contractor for this project shall: Remove old CAT 3 ethernet cables and re-wire 5 buildings on site and upgrade everything to current construction standards of CAT 6a. The total number of ports is approximately 26 with the main (internet source) building having 14 ports. The rest of the buildings have the current wiring supplied via underground conduits: Fish Culture Building (3 ports), Fish Health Office Building (5 ports), Virology/Necropsy (2 ports), and Maintenance (2 ports).

4.2 Contractor shall provide the following documentation:

- Network Diagrams: Detailed physical wiring diagrams showing the location of all cable runs, cabling routes.
- Port Inventory: A comprehensive list of all Ethernet ports, including their unique identifiers to be included with a LAN Room Cat6 Cable Test Report.
- Cable Labeling: Consistent and clear labeling of all Ethernet cables at both ends, aligning with the established documentation scheme.

5. Government-Furnished Equipment (GFE) / Government-Furnished Information (GFI)

5.1 Identify any Government-furnished equipment (GPE) and Government- furnished information (GFI): The equipment will be purchased, then US Fish and Wildlife Network Engineers can be on site to install the equipment

- Cisco C9200L Switch

- 3 - Meraki MR36 Wi-Fi 6 Indoor AP
- 3 - Meraki mGig 802.3bt PoE Injector
- 1 - 19dBi 120-degree antenna
- 2 - LTU Lite Client Radio
- 1 - LTU Rocket Base Radio
- 1 - MS 130 switch

6. Evaluation and Acceptance

6.1 The contractor will be evaluated as detailed herein. Government acceptance is contingent upon:

All new ethernet cables are installed correctly and terminated to industry standards. Cable testing results demonstrate compliance with the chosen cable category specifications for Cat 6a.

Network performance is enhanced, and connectivity issues are resolved.

Documentation is accurate and readily available.

7. Quality Control/Quality Assurance:

7.1 Identify and detail measures for assessing quality control and acceptance: A LAN Room Cat6 Cable Test Report to be provided at the completion of re-wiring.



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FOR IT INFRASTRUCTURE**

March 2024 Version 2.0

Document History:

Document Version	Date	Editor	Description of Revision(s)
2.0	March 2024	Jonathan Mitchum	Added to Table 1 • TIA-222 • TIA-322 • OSHA 1910.97 • Motorola R56 Modified the copper requirement for interbuilding runs. Included exterior building terminations for RF cables should be performed IAW Motorola R56, or as near as possible. Added terminations and connecting hardware, interbuilding entrance terminals (Circa Telecom 1880ENS1-25) to meet NEC. Modified guidance to reflect that 66 blocks are should only be used for cross-connect fields at Telco and interbuilding demarcs. Removed guidance that radio antenna leads shall not be installed in a TR/ER/EF. Added section for proper identification and labeling of RF cables. Added details for testing and reliability of RF cables.



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1.0 – General Information

Contractor

The Contractor shall provide and install a telecommunications distribution system meeting all applicable product and installation standards at time of installation. This system shall comprise: all entrance and inter-building cabling, telecommunications rooms (TR) and enclosures, equipment rooms (ER), entrance facilities (EF), backbone and horizontal cabling, terminations, and interfaces and, the associated connecting hardware, telecommunications infrastructure grounding and bonding, conduit, protective devices, firestop system, and other items as specified or required. This system shall support data applications.

The Contractor shall comply with the provisions of NEC, state codes, local codes, requirements of authorities having jurisdiction, manufacturer's requirements, the standards listed in Table 1: *Codes and Standards*, and these Specifications. The most current edition of codes/standards at time of installation applies. When there is a conflict the more stringent requirement applies.

If this construction is an office or building renovation, the contractor will be responsible for removal or demolition of the existing data and voice cabling from the ceiling and walls of the building. The old cabling creates an excess fire fuel load and, in some instances, may be a toxic hazard during a fire. This will decrease the liability of the U.S. Fish and Wildlife Service and the building owner in the event of a fire. Removal and demolition also reduce the weight on the cable trays and support system and decreases the chance of having the old cabling becoming a conductor for interference with the data and radio systems that remain in the office.

The Contractor provided telecommunications grounding and bonding system shall be installed by a licensed electrician.

Definitions

Telecommunications Architectural Spaces

Telecommunications Room (TR)

An enclosed space used exclusively for housing telecommunications equipment, cable terminations, cross-connection cabling, and interconnection to work areas. The TR is considered *floor serving*, opposed to building or campus-serving.

Any additional TR requirements specified elsewhere in this Agency Specific Requirements (ASR) shall contain the horizontal cross-connects to the main backbone cabling and to intermediate cross-connects to other TRs.

Telecommunications Enclosure (TE)

1. A case or housing, for telecommunications equipment, cable terminations, and cross-connect cabling.
2. A telecommunications space generally considered a *floor or tenant-serving* space smaller than a TR, providing a connection point between backbone and horizontal cabling.

Telecommunications Equipment Room (ER)

A centralized *building or campus serving* space used exclusively for telecommunications and/or computer equipment that usually houses a main or intermediate cross-connect.

Entrance Facilities (EF)

1. A secured, accessible entrance to a building for public and private network service cables, including connections/demarcation points to access provider, campus distribution, interexchange



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common carrier, the central station system for fire or burglar alarms, community antenna television (CATV) and closed-circuit television.

2. A *building or campus serving space/facility* that provides all necessary mechanical and electrical services, for the entry of telecommunications cables into a building. See 3.01

NOTE: In a **single** tenant building, any or all the EF, ER or TR functions may alternately be provided by a single room. These spaces may not contain a central station system for fire or burglar alarms, community antenna television, or closed-circuit television. These systems must be in a separate secure room as specified in ASR.

Demarcation Point

1. A point where the operational control or ownership changes.
2. The point of interface between access providers and customer facilities.

Network Interface (NI)

An FCC-registered device that provides an identifiable separation point between the access provider circuit and the premises switch or cabling.

Work Area (WA)

Building space where occupants interact with their telecommunications equipment, including systems furniture cubicles. The number of work area telecom outlet boxes shall be determined based on one (1) multi-port per fifty (50) square feet of rentable space.

Cabling Systems

The cables shall support voice and data systems in a single wiring grid (Do Not Separate voice and data cabling or patch panels). Optional cables will be required for radio and/or video if requested specifically in the ASR specifications.

Backbone Subsystem

Main backbone pathway or cable in a star topology providing interconnection between TRs, TEs, ERs, and the EF within or between buildings.

Horizontal Subsystem

Horizontal pathway or cable between and including the telecommunications outlet/connector in the work area and the horizontal cross-connect in the TR.

Firestop system

A specific installation consisting of fire-rated materials that fill the opening in a wall or floor assembly of, around and between any items that penetrate the wall or floor, and any termination devices along with their means of support.

Telecommunications Grounding and Bonding System

Grounds and bonds the telecommunications infrastructure. Complete definitions may be found in NEC and the ANSI/TIA-607-D Standard.

Telecommunications Main Grounding Busbar (TMGB)

A busbar placed in a convenient and accessible location and bonded, by means of the bonding conductor for telecommunications, to the building service equipment (power) ground and located near the building electrical service entrance.

Telecommunications Grounding Busbar (TGB)

A common point of connection for the telecommunications system and equipment bonding to ground and located in the telecommunications room or equipment room.

Telecommunications Bonding Backbone (TBB)

A conductor that interconnects the TMGB to the TGB.



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Telecommunications Infrastructure

1. A collection of telecommunications components, **excluding network and telephone equipment**, providing the basic support for the distribution of all information within a building or campus.
2. A substructure of a system used to support the cable system being installed.

Service Provider

1. A company (e.g., a telephone / internet company), that provides a circuit path between a service provider and the client user. An access provider can also be the service provider.
2. The operator of any facility that is used to convey telecommunications signals to and from customer premises.

Connecting Hardware

Devices providing mechanical cable terminations referring to cross-connects and mechanical connection hardware

Cross-Connects

A device enabling the termination of cable elements and their interconnection, and/or cross-connection, primarily by means of a patch cord or cross-connect jumper.

a. Main cross-connects (MC)

A cross-connect for first level backbone cables, entrance cables, and equipment cables.

b. Intermediate cross-connects (IC)

A cross-connect between first level and second level backbone cabling. Note: only one IC is allowed between a HC and the MC.

c. Horizontal cross-connect (HC)

A cross-connect of horizontal cabling to other cabling, e.g., horizontal, backbone, or equipment.

Pathway

1. The vertical and horizontal route of the telecommunications cable.
2. A facility for the placement of telecommunications cable.

Cable Trays and Wire/Pathways

Prefabricated structures used within commercial buildings for pathways of telecommunications cabling. These systems are placed and routed throughout all parts of the building, including access floors and ceiling pathways.

Copper Cable Over Voltage Surge Protection Devices

An over voltage protector featuring metallic electrodes that discharge in a gas atmosphere within glass or ceramic envelope. This type of protector does not require replacement each time it discharges.

Building Entrance Protector (BEP)

A device used for the termination, protection, and distribution of the cable pairs entering or leaving the building.

Codes and Standards

The following standards, as updated, are the applicable specifications for Communications Wiring and



FWS Agency Specific Requirements for IT Infrastructure

may be used for further clarification. The Contractor is responsible for knowledge of and adherence to the following applicable codes and standards:

Table 1: Codes and Standards

REFERRED TO AS	DOCUMENT TITLE
TIA-568-C.0	Generic Telecommunications Cabling for Customer Premises, Telecommunications Cabling System Structure, Cabling Installation Requirements, Cabling Transmission Performance & Test Requirements, Annex: A,B,C,D,E,& F.
TIA-568-C.1	Commercial Building Telecommunications Cabling Standard. ANSI/TIA-942 & ANSI/TIA-1005.
TIA-568-C.2	Balanced Twisted-Pair Telecommunications Cabling & Components Standard, Annex A,B,C,D,E,F,G,H,I,J,K,L, & M.
TIA-568-C.3	Optical Fiber Cabling Components Standard, Cable, Connecting Hardware, Patch cords & transitions (cassettes & breakout cables), Annex A.
TIA-569-C TIA-569-C-1 TIA-569-C-ERTA	Telecommunications Pathways and Spaces
TIA-606-B	Administration Standard for Telecommunications Infrastructure
TIA-607-B TIA-607-B-1 TIA-607-B-2	Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
TIA-455-C	General requirements for standard test procedures for optical fibers, cables, transducers, sensors, connecting and terminating devices, and other fiber optic components
TIA TSB-153	Static Discharge Between LAN and Data Terminal Equipment
TIA TSB-155-A	Guidelines for the Assessment and Mitigation of Installed Category 6 Cabling to Support 10GBASE-T
TIA TSB 162-A	Telecommunications Cabling Guidelines for Wireless Access Points
TIA/EIA TSB 184	Guidelines for supporting Power Delivery over Balanced Twisted Pair Cabling
TIA-758-B	Customer-Owned Outside Plant Telecommunications Infrastructure Standard
IEEE	The Institute of Electrical and Electronics Engineers
ANSI/TIA/EIA 598D	Optical Fiber Cable Color Code
NEC	National Electric Code, 2020 or latest version
NFPA	National Fire Protection Association
NESC	National Electric Safety Code, latest version
NEMA VE-1	National Electrical Manufacturers Association, Metal Cable Tray Standards, 1998
NEMA VE2-2001	National Electrical Manufacturers Association, Cable Tray Installation Guidelines
REA 345.52	Rural Electrification Agency Standard (January 16, 1980)
PE-89	Rural Electrification Agency Standard PE-89 (October 30, 1985)
ISO/IEC 8802-3	ANSI/IEEE 802.3 series of standards
BICSI TDM	Telecommunications Distribution Methods Manual, Eleventh Edition 2006 or latest Edition.
TIA-222 Rev I	Structural Standards for Antenna Supporting Structures, Antennas, and Small Wind Turbine Support Structures
TIA-322 Rev A	Loading, Analysis, and Design Criteria Related to the Installation, Alteration, and Maintenance of Communication Structures
OSHA 1910.97	Nonionizing Radiation
Motorola R56	Standards and Guidelines for Communications Sites



2.0 - Products

All materials and equipment furnished and installed shall be of the highest quality, new, and meet the standards of EIA/TIA, IEEE, UL, NFPA, and NEC and shall bear their label wherever standards have been established and label service is available.

Backboards

The backboards for the TRs, ER, EF termination devices shall be 20 mm (¾ in) thick, type ACX fire retardant plywood, covered with two coats of fire-retardant paint, and installed 8 inches AFF, 2.4 m (8ft) high.

Equipment Racks

Shall meet ANSI/EIA/TIA-310-D.

Open Frame

19" equipment rack, 72-84" high, with a flange base, standard channel depth, mounting rails drilled front and back and tapped to EIA standards.

- **Standard Solution:** Floor Mount – 2 Post – 45U
 - **Part Number:** SR2POST
 - **Dimensions:** Height: 84" Width: 22" Depth 14"

Quad Racks

Tapped to EIA standards.

- **Standard Solution:** Tripp-Lite Floor mount – Open- 42U
 - **Part Number:** SR42UBEXPND
 - **Dimensions:** Height: 78.5" Width: 24" Depth 43"

Quad Rack with Doors

Tapped to EIA standards.

- **Standard Solution:** Floor Mount – Enclosed – 42U
 - **Part Number:** SR42UB
 - **Dimensions:** Height 78.5" Width 24" Depth 45"

Wall Mounted Racks

- **Standard Solution:** Wall Mount – Enclosed – 15U
 - **Part Number:** SRW15US
 - **Dimensions:** Height: 31" Width: 24" Depth 22"

Equipment rack specifications shall be submitted by the Contractor for approval by the **Contracting Officer prior to purchase and installation.**

Grounding

Grounding Kit

Contains a busbar, busbar insulators and standoff brackets, and mounting equipment. Bolts shall be listed for grounding and bonding.

1. TMGB

Shall be predrilled copper, ¼" thick, 4" wide, and 20" long (minimum size).

2. TGB

Shall be predrilled copper, ¼" thick, 2" wide, and 12" long.



3. **Isolated Ground Bar**

Square D #PK15GTA-FT with a #PKGTAB insulator kit or equivalent.

4. **TBB**

Shall be insulated copper conductor.

Copper Wire and Cables

Cabling within ceilings used as a plenum for environmental air must conform to NEC fire ratings and be blue in color, or a color agreed upon by the Contracting Officer's Authorized Representative. The insulation and jacket types shall be CMP for Communications plenum, CMR or CMP for riser. Under-carpet wiring and flat wiring are not allowed.

Horizontal Cabling

New installations consist of 4-pair, Category 6A (CAT6A) or better, unshielded twisted pair (UTP), solid copper wires. Cabling to facilitate the connections to Wi-Fi access points in the ceiling or on the wall shall be the same as for all other Horizontal cabling.

Backbone Cabling

A copper inter and intra building backbone for digital applications, paging, analog applications requiring low voltage, and other miscellaneous uses.

Intra building

Shall be twenty-four 4-pair CAT6A or better cables terminated to a 24-port patch panel in the racks.

Under Ground Inter Building

Shall be 24AWG 25-pair direct cable (Superior Essex PE89-25P) for interbuilding runs less than 1000 feet, or 22AWG 25-pair direct burial cable (Superior Essex 20-062-05) for runs longer than 1000 feet.

Fiber Optic Cables

All fiber optic cables, equipment, and terminations shall comply with ANSI/EIA/TIA specifications. The insulation and jacketing for fiber optic cables shall be type OFN, OFNG for general wiring, type OFNP for plenum applications, and type OFNR or OFNP for riser applications. Cables, LC connectors, and equipment shall be approved for the location installed.

Horizontal Cabling

Fiber optic horizontal cabling, (from TR to WA) if required, will be identified in the ASR and shall consist of 6 pair 50/125 µm multimode OM4 or better fiber strands.

Backbone Cabling

Intra building

New installations shall consist of a minimum of twenty-four, 50/125 µm multimode fiber strands, unless otherwise specified in the ASR. During discussions, the Lessor and the FWS Telecommunications Subject Matter Expert shall discuss the type of cabling which will be required for networking equipment.

Under Ground Inter Building

New installations shall consist of a minimum of twenty-four, 50/125 µm single mode fiber or better fiber strands. Cable, corning (OSP) Outside plant cable loose tube pick filled, listed for direct burial cable. Can be installed in non-conductive conduit but can only have two 90-degree sweeps. If more turns are required, Hand Holds (HH) or



Manholes (MH) must be installed in accordance with the BICSI TDMM. Termination of Fiber Cable will be in 19-inch rack mountable LIU's, using LC connectors.

Exterior Building Terminations

Exterior Wall

Two 4-pair CAT6A or better cables shall be installed from the patch panel to the exterior termination point in the building. Termination on the exterior wall shall consist of commercial-grade stainless steel water-resistant wall plate and Cat6A commercial-grade outdoor shielded panel mount keystone jack with dust cap.

Exterior RF Terminations

Exterior Building Terminations for RF cables should be performed IAW Motorola R56, or as near as possible. Project Managers should consult with the Radio Team for site specific requirements.

Wireless Bridge

Two 4-pair CAT6A or better cables shall be installed from the patch panel to the exterior termination point in the building. Termination on the exterior wall shall consist of commercial-grade stainless steel water-resistant wall plate and Cat6A commercial-grade outdoor shielded panel mount keystone jack with dust cap. Both cables shall be installed with an ethernet surge protector that is grounded and installed by a certified electrician.

Exterior Post

Shall be CAT6A outside plant (OSP) type cable, shielded, twisted pair, 24 AWG solid copper, moisture resistant filled and listed for direct burial or installation in underground conduit. Both cables shall be installed with an ethernet surge protector that is grounded and installed by a certified electrician. Termination shall be specified by the Contracting Officer to meet the individual application needs.

Terminations and Connecting Hardware

Intra Building Cable Protector

Shall meet the specifications of the Building Entrance Protector (BEP), or equivalent.

Under Ground Inter Building Entrance Terminals

Circa Telecom 1880ENS1-25 to meet NEC.

Copper Cable Over Voltage Surge Protection Devices

Industry standard 5-pin gas or solid-state modules that fit the BEP.

66 Block

CAT6A or better, 25 or 50 pair capacity with stand-off brackets. These should only be used for cross-connect fields at Telco and interbuilding DEMARCs.

110 Block

CAT6A or better, T568A wiring, with jacks.

CAT6A Patch panels

Shall be standard 19-inch rack mountable modular T568A, minimum 24 port and maximum 48 port; and include front, side, and rear cable management, icon label holders, designation labels, cable ties, mounting hardware.

- **Standard 48 Port Rack Mount Solution:** Tripp-Lite – Patch Panel- 48 Port



- Part Number: N252-048
- **Standard 24 Port Rack Mount Solution:** Tripp-Lite – Patch panel – 24 Port
 - Part Number: N252-024

Horizontal Cable Management Panels

Must be approved by Contracting Officer.

Fiber Optic connectors

Shall be LC style connectors.

Fiber Optic LIU

Shall be standard 19" rack mountable with LC style connectors, fiber managers, port designation labels, strain relief hardware, cable ties, and mounting hardware.

Fiber Optic Wall Mount Unit

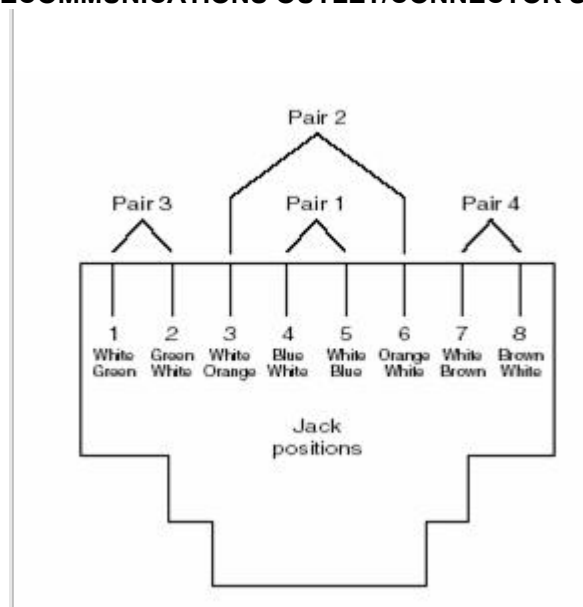
A wall mounted unit may be used if approved by the Government.

Work Area/Wi-Fi Telecommunications Outlet/Connectors

Work Area (WA) and Wi-Fi telecom outlet/connectors shall be CAT6A, 8-pin modular T568A (Figure 1: Telecomm Outlet/Connector Specifications). Jack color shall be consistent with other plate colors, icon colors shall be coded to represent the following applications (or colors agreed upon by the Contracting Officer's Authorized Representative):

- Government Data – Blue
- Radio – Yellow
- Public Data – Red

FIGURE 1: TELECOMMUNICATIONS OUTLET/CONNECTOR SPECIFICATIONS



Eight Position Jack Pin/Pair Assignments
Illustration is a front view of the connector
T568A

COLOR CODES



IRTM Information Resources and Technology Management

FWS Agency Specific Requirements for IT Infrastructure

<u>Conductor</u>	<u>Color Code</u>	<u>Abbreviation</u>	<u>Pin Number</u>
Pair 1 (NOTE 1)	White-Blue Blue	W-BL BL	5 4
Pair 2 (NOTE 1)	White-Orange Orange	W-OR OR	3 6
Pair 3 (NOTE 1)	White-Green Green	W-GR GR	1 2
Pair 4 (NOTE 1)	White-Brown Brown	W-BR BR	7 8

NOTE 1: A white marking is optional.

Outlet Box

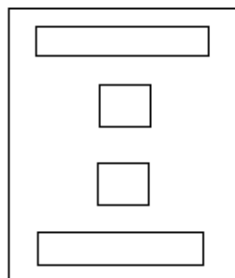
Wall outlet boxes shall be no smaller than 50 mm (2") wide, 75 mm (3") high, 64 mm (3½") deep, and have the capability to accommodate one or two 21 mm (¾") trade size conduits.

Work Area Faceplates

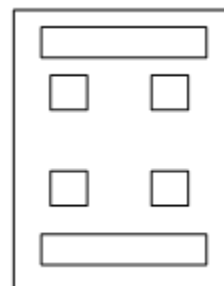
Wall faceplates shall fit the installed outlet box, 2-4 ports, flush mountable, with label designations and covers, icons, and dust covers for unused ports.

FIGURE 2:

2 PORT FACEPLATE CABLE LOCATION



4 PORT FACEPLATE CABLE LOCATION



Training, Conference, Interview and Equipment Rooms

Faceplates shall fit the installed outlet box, 2 port, flush mountable, with label designations and covers, with an option for dust covers for unused ports.

Systems Furniture Faceplates

Shall fit the furniture. The contractor shall coordinate with the locations systems furniture point of contact to determine if faceplates are included. The contractor shall provide the faceplates if they are not part of the systems furniture package and shall be the type recommended by the systems furniture vendor.

Cross-Connect / Patch Cords

Government shall supply copper and fiber patch cords.



FWS Agency Specific Requirements for IT Infrastructure

1. Required in the TRs to cross connect the voice/data equipment to the appropriate patch panel
2. Required in the WA between the outlet box and equipment.

Wireways

Conduit

Recommended conduit: Intermediate Metal Conduit (IMC), Rigid Metal Conduit (RMC), Rigid Nonmetallic Conduit (RNC), Nonmetallic Underground Conduit with Conductors (NUCC, Electrical Metallic Tubing (EMT), Electrical Non-Metallic Tubing (ENT). Refer to Section 3.08 D. Conduit Sizing for requirements.

Cable Trays

Includes listed factory manufactured tees, crosses, risers, elbows and other fittings of the same material as the main sections. Materials may be any of the following:

1. Galvanized steel
2. Steel with factory applied paint
3. Aluminum alloy 6063-T6 with compatible alloy for parts

Wire Basket Support System

Listed straight sections of continuous wire mesh, field formed horizontal and vertical bends, tees, dropouts, supports and accessories. Cooper B-line Wire Basket Runway or equivalent.

Innerduct Tubing

HDPE (high density polyethylene) designed for inside building installation of fiber-optic cables.

Wire Management Hardware

Cable hooks, wire ties and mounting bases, brackets, clips, and other similar devices to provide support or bundling for cables shall be screw or otherwise permanently mounted. Glue mounted devices are not allowed.

Firestopping

Selected systems shall meet the hourly time delay ratings recommended for each fire-rated floor, wall, or other partition of building constructions; and be appropriate for each type of telecommunications penetrations per NEC and the additional requirements in ANSI-J-STD-607-A.

3.0 - Execution

Entrance Facility

One Entrance Facility (EF) is required per site. If the entrance facility is separate from the ER, it shall be 8'x 8' minimum, and sized upward according to the applications supported according to TIA-569-C and meet the TR requirements. If office space is multi-tenants, then a single EF must be provided and separate from the Government's ER.

Service Provider/Telephone Company Interface – The Contractor is required to coordinate, with the Service provider, copper telecommunication line installation, termination and identification. The copper telecom lines shall be terminated on a BEP if required by code.



FWS Agency Specific Requirements for IT Infrastructure

Access to the building electrical ground must be provided, by the contractor, within 1.5 (5 ft) of the OSP conductive cable terminations in the EF. The access provider will make cable metal sheath (bond and ground) connections to the provided electrical ground.

Equipment Room

Recommended 10 ft x 11 ft and capable of serving different types of telecommunications applications and include space for environmental control equipment, power distribution/containers, and uninterruptible power supply (UPS) system, and accommodate the telecommunications rack and equipment clearance space requirements. This space requirement does not include the footprint of the door that must swing inward to secure the hinge pins. ER sizes are specified in the design phase. If this room is sized in the SFO, the sizing in the special requirements section will override this section.

Horizontal, backbone, and entrance wiring systems; patch panels; equipment racks; and cross-connect blocks shall be arranged logically in the ER. Allow for natural wiring progression, and growth, minimal wire crossing, and easy access to each component for testing and facilitating moves, additions, and changes.

Equipment racks must be bolted to the floor by means of Unistrut or concrete lag bolts and shall have ladder rack raceway anchored to the top of the equipment rack with a bracing kit and anchored to the wall(s) with a wall angle support kit. Allow for 40" wide, 40" depth, and 7.5 ft high for each equipment rack. Provide space for an aisle of at least 3 ft wide in the front and in the rear of the space allocated for each equipment rack or cabinet. Equipment rack mounting must meet seismic requirements.

The telecommunications connecting hardware shall be rack or wall mounted according to individual site requirements, see Section 2.

Multi-Tenant Buildings

Government TR/ER shall not be shared with other tenants or federal partners or used for any purpose other than a telecommunications facility.

Proximity to Electrical Power Service and EMI Sources

Equipment with potential sources of EMI shall be located no closer than 10 ft to the TR/ER.

Heating, Ventilating, and Air Conditioning (HVAC)

Shall be installed and function properly 24 hours per day, 365 days per year, temperature 64° F to 75° F, and relative humidity 30% to 55% connected to the emergency power source, if required. See requirements elsewhere in this Agency Specific Requirements (ASR) for the BTUs. HVAC sensors shall be in the ER and placed 1.5 m (5 ft) AFF.

Ceiling Height

3 m (10 ft) above finished floor, no false ceilings.

Doorways

Install fully opening, lockable, removable doors that are at least 3.0 ft wide and 7.0 ft tall. The hinges shall be so that the pins are inside the TR door. When necessary, double doors are to be installed side to side and be removable. Door sills and center posts are not allowed.

Floor Requirements

Shall be non-conductive vinyl or ceramic tile and shall be installed with the appropriate non-conductive adhesive and grout. Floor loading TR: a minimum 50 lbf/ft². Floor loading ER: a minimum of 100 lbf/ft² to 250 lbf/ft².



Electrical Power

Follow the latest NEC code and reference BICSI TDM, include provisions for UPS power draw during recharge cycles. Consult a qualified electrical engineer for design and implementation.

Backboards

Shall be installed 8" AFF, with C side facing the wall. Support devices are required above and below termination devices to support the cross-connects. Install a TGB Ground Kit on one of the backboards in each TR. (See [Section 3.08 Protection from Physical Damage, sub sections: B. Pathways and C. Required Conduits.](#))

Equipment Racks

The telecommunications connecting hardware shall be floor or wall rack mounted according to individual site requirements, see section 2.03. Equipment rack mounting must meet seismic conditions.

Floor mount equipment racks must be bolted to the floor by means of Unistrut or concrete lag bolts and shall have a 12" ladder rack cable tray; with 12" rung spacing; anchored to the top of the equipment rack with a bracing kit and anchored to the wall(s) with a wall angle support kit. Allow for 40" wide, 40" depth, and 7.5 ft high for each equipment rack.

Provide space for an aisle of at least 3 ft wide in the front and in the rear of the space allocated for each floor mounted equipment rack or cabinet. Provide space for an aisle of at least 3 ft wide measured when the rack is opened to allow access to the rear of each wall mounted equipment rack.

Telecommunications Rooms

TRs shall be designed and located so that no wire run from a TR to a work area telecom outlet is more than 90 meters (**295 ft**). Contractor located TRs shall be in the core of the building, adjacent to, or in the core of the area served by that TR. The room requirements shall meet Section 3.02. Climate control shall be to standard office temperatures.

Telecommunications Room Sizing/Number Required

TRs shall be provided on each floor unless otherwise permitted, according to Table 2: *TR Sizing*. Additional TRs shall be provided when the floor area served is greater than 1000 m² (10,000 sf), or the distribution distance to the work area exceeds 90 m (295 ft).

Service Area		Closet Size	
m ²	ft ²	mm	ft
1000	10,000	3000 x 3400	10 x 11
800	8000	3000 x 2800	10 x 9
500	5000	3000 x 2200	10 x 7
under 500 (small buildings only)	< 5000	2440 x 2440 min. walk-in closet	8' x 8' minimum

Table 2: TR Sizing
(EIA/TIA 569 Table 7.2-1 Modified)
Maintain working space and clearances per NEC.

Grounding

Grounding shall be in accordance with NEC, local codes, and the additional requirements in ANSI/TIA-



607-D and installed by a licensed electrician. When there is a code conflict the most stringent requirement shall be followed.

TGMB

Install within 5 feet of the main electrical service of any building with more than one TR. Bond to the electrical service with a TBB. The TGMB and TBB's shall be protected from damage, tampering and the weather.

TGB

Install one in each TR on one of the backboards. Ground to the TGMB if installed for this building or directly to the main electrical service for the building when a TGMB is not installed with a TBB.

TBB

Minimum #6 AWG up to maximum 3/0AWG, size according to NEC grounding electrode conductors and any additional requirements in TIA-607-B.

Where there is a load center co-located in the same room as telecom equipment, this load center shall be bonded to the ground bar in addition to the load center's normal grounding path.

All protection devices, frames, racks, cabinets, and telephone and data equipment shall be bonded to the TGB according to NEC, and the additional requirements in TIA-607-D. When compatible or more stringent than code, follow grounding instructions of the equipment manufacturer.

Grounding paths to the ground bar shall be kept short, (under 1 meter (3.3 ft), install additional isolated ground bars as needed). **Do not form ground loops.**

Copper Wire and Cables

Horizontal Cables

All wiring and wiring component installation shall meet the requirements of ANSI/TIA/EIA and BICSI TDM, CAT6A installation procedures, and all applicable manufacturer installation requirements. The maximum cable length between the TR to WA telecom outlet box shall not exceed 90 m (295 ft). Cabling to facilitate the connections to Wi-Fi access points in the ceiling or on the wall shall be the same as for all other Horizontal cabling. Refer to Section 3.09 Identification, Labeling, and Documentation Requirements for labeling instructions.

Two (2) continuous (no splicing allowed), horizontal cables shall be installed to each WA and Wi-Fi telecommunications outlet box from the serving TR.

Voice, radio, and data cables shall not be placed in the same conduit or raceway with electrical power distribution components according to NEC.

Each WA or Wi-Fi cable shall be terminated on a separate T568A telecommunications outlet/connector, and on the serving TR, CAT6A patch panel, at the other end.

Work Area Telecommunications Wall Outlet Box

Shall be located as identified in the design layout. The work area telecommunications outlet box should be located within 1 m [3 ft] of an electrical outlet and installed at the same height if appropriate. The telecom outlet boxes shall be located and recessed in the wall, floor, or power pole so that a faceplate can be flush mounted. Wi-Fi outlet boxes can be mounted above the drop ceiling or flush with the ceiling tiles or hard ceiling.



FWS Agency Specific Requirements for IT Infrastructure

Furniture Systems Horizontal Pathways

The contractor shall install all wire and cables in furniture systems per EIA/TIA-569-C, Furniture Pathways and Spaces. A minimum of Two (2) CAT6A connections are required in each workstation.

The Government will install all cross-connect cables for voice, fax, radio and data application.

Backbone Cable

Intra Building

Install a minimum of twenty-four 4 pair CAT6A cables and terminate them on a 24-port patch panel or greater. Backbone cables shall extend from the ER to TRs in a physical star topology. Each TR is wired to the ER main cross-connect or to an intermediate cross-connect then to a main cross-connect. No more than two hierarchical levels of cross-connects are allowed in the backbone wiring. CAT6A cables to be used for local area network connections shall terminate on CAT6A patch panels mounted on an equipment rack. CAT6A cables to be used for other digital applications, paging applications, analog applications requiring low voltage shall be terminated on a 66 block, with stand offs.

Underground Inter Building

Install one approved 24AWG 25-pair direct cable (Superior Essex PE89-25P) for interbuilding runs less than 1000 feet, or one approved 22AWG 25-pair direct burial cable (Superior Essex 20-062-05) for runs longer than 1000 feet. The cable pairs must be terminated at each end on a Circa Telecom 1880ENS1-25 to meet NEC. Install the over voltage surge protection device for each pair entering or leaving the EF.

Fiber Optic Backbone Cable

Install fiber optic backbone cables for inter-building and for all intra-building connections between TR's, ER and EF.

Fiber optic, backbone cables shall extend from the ER in a physical star topology. Each TR is connected to the ER via a main cross-connect or an intermediate cross-connect then to a main cross-connect. No more than two hierarchical levels of cross-connects are allowed in the backbone cabling.

Terminations

Install specified connectors on all fiber strands when installing fiber optic cable.

Light interface units (LIU) or termination enclosures shall be installed at each TR as required for terminations. The recommended 3 m (10 ft) slack for fiber optic cables will be stored in an extended loop or in a figure-eight.

The Government will install all fiber optic cross-connect cables.

Protection from Physical Damage

The cable network shall be structured and equipped in accordance with NEC and EIA Standards to minimize vulnerability. All wire and cable shall be routed away from potential sources of mechanical and electrical damage or interference. All exposed interior and exterior cables shall be protected from physical damage by installing it in the conduit type required by NEC.



FWS Agency Specific Requirements for IT Infrastructure

Support

Telecommunications wiring shall be secured and supported using cable trays or J-hooks. If J-hooks are used, the cabling shall be supported between 48 and 60 inches off-center. The cabling will be supported not more than 6 inches (150 mm) from equipment racks, frames, and terminals. Attic joists, suspended ceiling panels, duct work and other similar items are not allowed for cable support.

Support devices are required above and below termination devices to support the cross-connects.

These support systems shall contain only telecommunication cable; electrical cable is not allowed to be placed in the same systems.

Pathways

Provide cable trays or wire basket support systems above the ceiling or below the floor for telephone and data cabling for all main corridors (**implied corridors in open office areas and/or partitioned corridors**), i.e., from TRs and ERs to the work areas and sized in accordance with NEC, TIA 568-C and TIA-569-C specs and NEMA VE-2. The system is for exclusive use by Government and shared partners. The type and location of wire ways shall be determined during the building design in accordance with NEC and the additional requirements of NEMA VE-1.

The tray/basket type support capacity shall be determined in accordance with the manufacturer's maximum recommended load capacity. A 50% fill capacity shall not be exceeded, and cables shall be stacked no higher than 6".

If the pathway is in the ceiling, allow for 12" of clear vertical space above the tray/basket type systems and 6 inches between the suspended ceiling and the cabling pathways. Ceiling tiles shall not be placed higher than 11 ft AFF. If the building does not have a T-bar grid suspended acoustical ceiling with access to plenum and trays, a minimum of one access point per room must be provided.

Bond metal pathway systems to the telecommunications grounding and bonding system. Space at least 12" from trays of other systems. Access must be provided between floors for the connection of the support system. The horizontal pathway placement shall be coordinated between the various trades that use the space.

Pathways other than main corridors (auxiliary runs) – When using J-hooks, the internal diameter shall be no larger than 2 inches. Installation and fill ratio must adhere to manufacturer's guidelines.

Required Conduits

Off premise entrance conduit shall include at least two 100 mm (4 in) RSC per building stubbed out from the EF for telephone entrance, etc., NEC Section 8. Sharing entrance facility conduit with separate providers is not allowed.

Provided the cables and radio equipment are installed correctly and the cables have adequate separation (min 12" is typical), Radio frequency (RF) interference should not occur. Radio antenna lead locations will be identified in the ASR.

Innerduct tubing shall be installed in each backbone pathway where fiber optic is used and sized to accommodate the BICSI TDM recommended fill capacity.

A ¾" conduit minimum shall run from each telecom work area outlet box, through the wall, to



accessible locations above the ceiling or below the floor. The conduit shall terminate horizontally, above or below all obstructions and readily accessible, directed toward the horizontal pathway. Size this conduit for the installed horizontal cables plus 2 spare uninstalled cables.

All exposed cables shall be protected from physical damage by installing in an approved wire way.

Conduit Sizing

Backbone and inter building conduits shall be sized to include 25 to 30 percent spare capacity for future cables. Conduits shall be sized according to ANSI/TIA/EIA 569 as depicted in Table 3: *Conduit Sizing*, unless a larger size is specified. Minimum 4" conduit required between buildings.

Conduit			Number of Cables or Wires									
Internal Diameter		Trade Size	Wire O.D. mm (in)									
mm	(in)		3.3 (.13)	4.6 (.18)	5.6 (.22)	6.1 (.24)	7.4 (.29)	7.9 (.31)	9.4 (.37)	13.5 (.53)	15.8 (.62)	17.8 (.70)
15.8	0.62	1/2	1	1	0	0	0	0	0	0	0	0
20.9	1.82	3/4	6	5	4	3	2	2	1	0	0	0
26.6	1.05	1	8	8	7	6	3	3	2	1	0	0
35.1	1.38	1-1/4	16	14	12	10	6	4	3	1	1	1
40.9	1.61	1-1/2	20	18	16	15	7	6	4	2	1	1
52.5	2.07	2	30	26	22	20	14	12	7	4	3	2
62.7	2.47	2-1/2	45	40	36	30	17	14	12	6	3	3
77.9	3.07	3	70	60	50	40	20	20	17	7	6	6
90.1	3.55	3-1/2	-	-	-	-	-	-	22	12	7	6
102.3	4.02	4	-	-	-	-	-	-	30	14	12	7

**Table 3: Conduit Sizing
(TIA 569-C)**

Conduit Installation

All conduit installations shall meet NEC and the additional requirements in the BICSI TDM manual and shall be bonded and grounded according to ANSI J-STD-607-A. Changes in direction shall be sweeping bends or pull boxes. Maximum of 30 m (100 ft) between pull boxes where installed above ground.

Minimum burial depths shall be per NEC Chapter 3. A pull cord shall be installed in all backbone and innerduct tubing.

Identification, Labeling, And Documentation Requirements

A numbering system shall be used for the wiring, in which a single sequence of numbers does not serve more than one office number. This numbering system shall consist of a unique telecommunications outlet/connector numbering plan. The numbering system shall be consistent throughout the building or campus.



FWS Agency Specific Requirements for IT Infrastructure

All termination devices shall be stenciled and labeled in accordance with ANSI/TIA/EIA606-A standard.

The horizontal cable from the telecom outlet box to the TR shall be labeled with identical information at each end.

RF cables shall be marked with rings of 3-4 wraps of colored ¾" electrical tape (e.g. Scotch 35). Marking tape shall be applied within 6" of the indoor terminal connector and every 10 feet thereafter. If the final mark is more than 3 feet from the outdoor terminal connector, a mark shall be applied within 6" of that connector, as well. Marking colors shall be Red for VHF band, Yellow for UHF band, Blue for 800MHz band, and Green for any frequency above 1GHz. In cases where more than one cable for a band exists, additional rings of the appropriate color spaced 2" apart shall be applied (2 for second cable, 3 for third, etc.) Cables shall also be tagged with information for their associated antennas. Tags shall contain the antenna type and location on the tower (side/leg & height).

Upon completion of the installation, wire and cable plans shall be prepared and posted in each Telecommunications Room. Within thirty (30) days after occupancy, Contractor must provide the Contracting Officer with as-built drawings to a scale of 1/8" equals one foot on via digital means (flash drive, email) to the Contracting Officer. As built drawings must identify all wire and cable, including cable tray layouts.

Testing and Reliability

The contractor shall test the system in the presence of the Contracting Officer or other agency representatives. Tests shall be made of each cable segment with copper and fiber optic certifying devices (tester), a printout of each cable segment shall be given to the Contracting Officer. All certification data will be provided to the Contracting Officer digitally (flash drive, email).

The CAT6A cable system shall be certified to meet the ANSI/TIA/EIA568C.0, Transmission Performance Specifications for 4-pair CAT6A cabling. All grounds, opens, shorts or other cable or connecting component defects shall be rectified and certified at no extra cost to the agency before acceptance.

Fiber Optic cable shall be certified to meet the Fiber Optic Test Procedure (FOTP) Standards developed and published by the Electronic Industries Association (EIA) under the EIA's RS-455 series of standards, for the type of fiber optic cable installed.

RF cables shall be tested with a suitable antenna analyzer (e.g. Bird Sitehawk, Anritsu Site Master). Markers shall be set at site specific frequencies and measurements displaying either SWR or Return Loss shall be captured by screenshot or suitable jpeg output file showing the waveform and the values at each marker. Acceptable measurements shall be ≤ 1.5:1 for SWR or ≤ -14dB for Return Loss. Any cable not passing acceptance shall be corrected. All certification data will be provided to the Contracting Officer digitally (flash drive, email).

Firestopping

The methods, materials, and considerations for reestablishing the integrity of fire-rated architectural structures and assemblies, required by building codes must be observed when these barriers are penetrated by cables, pathways, or other penetrating elements.

Coordinate each firestop selection with adjacent work for dimensional or other interference and for feasibility. In areas accessible to the public and other finished areas, firestop systems work shall be selected, installed, and finished to the quality of adjacent surfaces of building construction being penetrated.

Use materials without irritating or objectionable odors when firestopping is required in existing buildings and areas that are occupied.



FWS Agency Specific Requirements for IT Infrastructure

Provide damming materials, plates, wires, restricting collars, and devices necessary for the installation of firestopping. Remove combustible installation aids after firestopping material has cured.

All firestops shall be installed in accordance with the manufacturer's instructions in order to maintain the specific rating assigned.

Firestopping must be inspected by the local Fire Marshall and verified by the Contracting Officer or other designated Government representative. If firestopping is exposed for inspection as directed by the inspecting authority to permit his or her inspection, reinstall new firestopping and restore work where removed for inspection. This requirement exists because firestopping is often installed when an inspecting official is not available.





Information Technology Baseline Compliance Contract Guidelines

July 2022

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Introduction

This document provides guidance regarding requirements that must be reviewed and considered for incorporation, where applicable, in all Department of the Interior (DOI, Department) contract actions that involve information technology (IT) products or services. Contracting Officer Representatives (CORs) and/or the requirement owners should work closely with their Bureau or Office Associate Chief Information Officer and Contracting Officer (CO) to ensure the below requirements are properly addressed in IT contracts within the appropriate sections. Any referenced regulation or policy should be provided to the Contractor via hypertext link or other means.

Section 1.0 – Information Technology and Cybersecurity Requirements

A. Applicability

This term applies to the Contractor, its subcontractors, and Contractor employees (hereafter referred to collectively as “Contractor”). The Contractor shall use these terms in all subcontracts.

B. Requirements

Information systems and system services provided to the Department by the Contractor must comply with DOI IT Cybersecurity and Privacy Control Standards, privacy policies, and other related guidance. Contracts that involve Industrial Control System (ICS) products and services should be evaluated on an individual basis and contain language specific to ICS. Please contact the Bureau of Reclamation (BOR) at ICSsecurity@usbr.gov for ICS issues and guidance.

The Contractor and the contractor-managed systems shall comply with all federal regulations and guidance, including but not limited to:

1. Federal Information Security Modernization Act of 2014 (FISMA) accessible at <https://www.cisa.gov/federal-information-security-modernization-act>
2. Federal Risk and Authorization Management Program (FedRAMP) for cloud hosted systems accessible at <https://www.fedramp.gov>
3. Section 208 of the E-Government Act of 2002 accessible at <https://www.govinfo.gov/content/pkg/PLAW-107publ347/pdf/PLAW-107publ347.pdf>
4. Privacy Act of 1974 accessible at <https://www.justice.gov/opcl/privacy-act-1974>
5. Federal Information Processing Standards (FIPS) and the National Institute of Standards and Technology (NIST) 800-Series Special Publications (SP) accessible at <https://www.nist.gov>
6. Office of Management and Budget (OMB) Circular A-130, *Managing Information as a Strategic Resource*, accessible at <https://www.cio.gov/policies-and-priorities/circular-a-130/>

C. Information Systems

The Contractor shall manage information in accordance with applicable laws, regulations, executive orders, and policies regarding the marking, safeguarding, and dissemination of

Controlled Unclassified Information (CUI). Personally Identifiable Information (PII) is a subset of information designated as CUI, and Sensitive PII is a subset of PII that requires additional controls and safeguards. (See section 3, Privacy Requirements)

The Contractor shall not use or redistribute any sensitive information processed, stored, and/or transmitted by the Contractor except as specified in the contract.

Contractor Information Systems shall comply with FISMA. The Contractor shall provide an Assessment and Authorization (A&A) documentation package for each system that complies with Federal Information Processing Standard (FIPS) 199, National Institute of Standards and Technology (NIST) Special Publication (SP) 800-34, NIST SP 800-37, NIST SP 800-53, NIST SP 800-60, et al.

The Contractor cryptographic modules used to protect DOI information shall comply with the current NIST FIPS 140 version and be validated by the Cryptographic Module Validation Program.

Access to all government data stored outside the confines of a DOI-managed data center shall comply with Trusted Internet Connections (TIC) requirements.

All Contractor personnel requiring access to DOI information data and/or systems shall be cleared at appropriate levels as set forth:

1. Low Risk position will require a National Agency Check with Written Inquiries (NACI) or equivalent investigation.
2. Moderate Risk position will require either a Limited Background Investigation (LBI) or a Minimum Background Investigation (MBI).
3. High Risk position will require a Background Investigation (BI).

Vulnerability Scanning - All IT systems providing services to or on behalf of the Department or storing, processing, or routing DOI CUI information must be scanned monthly with a vulnerability analysis tool using authenticated scans that are acceptable to DOI. Vulnerabilities and weaknesses shall be remediated in accordance with (IAW) Department-mandated time frames commensurate with the level of risk.

D. Cloud

All Contractor cloud-hosted Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS) or other “as a Service” offerings shall comply with applicable language within the DOI cloud hosting program guidelines.

E. Internet Protocol Version 6 (IPv6)

The applicability of IPv6 to DOI-related components (networks, infrastructure, and applications) specific to individual acquisitions shall be in accordance with the Department Acquisition, Arts, and Asset Policy DOI-AAAP-0055, *Procuring Internet Protocol Version 6 (IPv6)*, (per OMB Memorandum M-05-22, August 2, 2005) and with subsequent federal requirements (OMB Memorandum M 21-07 *Transition to IPv6*, September 28, 2010) regardless of whether the

acquisition is for modification, upgrade, or replacement. All Enterprise Architecture related component acquisitions shall be IPv6-compliant as defined in the U.S. Government Version 6 (USGv6) Profile (NIST SP 500-267) and the corresponding declarations of conformance defined in the USGv6 Test Program. In cloud service procurements, IPv6 compliance should be pursued where practicable.

F. Continuous Monitoring

All Contractor-operated systems that input, store, process, output, and/or transmit DOI CUI shall meet or exceed the continuous monitoring requirements identified in the DOI Continuous Monitoring Plan or the Bureau or Office Continuous Monitoring Plan, whichever best fits mission, security, and risk management needs.

Contractor-managed systems are required to have an assigned Information System Security Officer (ISSO) and System Owner (SO) to support the processes described in the contract. While the procuring organization designates government personnel to serve in these roles, the contractor must provide functionally equivalent personnel to enable effective day-to-day operation, management, and monitoring of the system. The Contractor ISSO and SO resources are required to comply with DOI- or other-approved annual and Role Based Security Training (RBST) requirements.

Contractors managing systems on behalf of the Department are required to submit Continuous Monitoring Reports per the Continuous Monitoring Plan to the federal point of contact designated in the contract per the timelines prescribed in the Continuous Monitoring Plan. The Contractor-managed systems may be required to submit to a penetration test (at the discretion of the awarding program) in coordination with DOI cybersecurity personnel or approved contract officials.

When required by the contract, the Contractor must provide the following in alignment with DOI's continuous monitoring strategy:

1. account management
2. vulnerability management
3. patch management
4. antivirus
5. malware detection
6. event management
7. configuration management
8. license management
9. incident management

G. Security Training and Reviews

All users of federal information systems are required by Title 5, Code of Federal Regulations, Part 930.301, as amended, to be exposed to security awareness materials annually or whenever system security changes occur, or when the user's responsibilities change. The Department requires that Contractor employees take an annual Information Technology Security Awareness Training course before accessing federal information systems under the contract. Unless otherwise specified, the training shall be completed within thirty (30) days of contract award and

be completed on an annual basis thereafter. New Contractor employees assigned to the contract shall complete the training before accessing federal information systems under the contract. The Contractor shall maintain copies of training certificates for all Contractor and subcontractor employees as a record of compliance. Unless otherwise specified, initial training certificates for each Contractor and subcontractor employee shall be provided to the COR not later than thirty (30) days after contract award. Subsequent training certificates to satisfy the annual training requirement shall be submitted to the COR via e-mail notification each year. The e-mail notification shall state that all Contractor and subcontractor employees have completed the required training.

The government may elect to conduct random periodic reviews to ensure that the security requirements contained in this contract are being implemented and enforced, particularly when contractors build and operate or maintain systems on behalf of the Department. In such cases, the Contractor shall afford DOI, the Office of the Inspector General, and other government organizations access to the Contractor's facilities, installations, operations, documentation, databases, and personnel used in the performance of this contract. The Contractor shall provide access, to the extent necessary as determined by the government, for the government to carry out a program of inspection, investigation, and audit to safeguard against threats and hazards to the integrity, availability, and confidentiality of government data or the function of computer systems used in performance of the contract and to preserve evidence of computer crime. Cloud Service Providers should not be expected to grant physical access to government inspectors.

H. Contractor Reporting Requirements

The Contractors operating information systems on behalf of the Department shall comply with FISMA reporting requirements. Department system owners shall coordinate annual and quarterly data collection. Contractors must provide DOI with the requested information based on the timeframes provided with each request. The Contractor systems shall comply with near real time feeds in accordance with Department of Homeland Security (DHS) Continuous Diagnostics and Mitigation (CDM) requirements as coordinated by DOI. Reporting requirements are determined by OMB and may change each reporting period.

The Contractor-managed systems shall adhere to NIST 800-53 and submit to an annual assessment performed in coordination with DOI cybersecurity personnel or approved contract officials. All results of the annual assessment become the property of the Department.

The Contractor shall comply with DHS or other government oversight bodies regarding cybersecurity-related mandates and directives (Binding Operational Directives [BOD], etc.) or any additional ad-hoc reporting requirements.

Section 2.0 – Cybersecurity Incident Response

A. Applicability

This term applies to the Contractor, its subcontractors, and Contractor employees (hereafter referred to collectively as "Contractor"). The Contractor shall use these terms in all subcontracts.

B. Definitions.

As used in this term-

“Incident” means an occurrence that

1. actually or imminently jeopardizes, without lawful authority, the integrity, confidentiality, or availability of information or an information system; or
2. constitutes a violation or imminent threat of violation of law, security policies, security procedures, or acceptable use policies.

C. Requirements

All Contractor employees and information system service providers shall be responsible for recognizing and reporting security incidents. Information system service providers shall receive the current version of the DOI Incident Response Plan, Policy, and Handbook at the time of award, in addition to bureau- or office-specific guidance for appropriate reporting and response at all levels. All suspected or confirmed information security incidents shall be reported in accordance with the requirements listed below, even if it is believed the incident may be limited, small, or insignificant. For hosting environments, including cloud systems and services, this includes incidents occurring on the service provider’s underlying infrastructure supporting containers utilized by the Department. In such cases, the vendor’s primary security operations and/or incident response unit shall engage directly with the DOI Cyber Security Operations (CSO) Section and provide DOI CSO with security and event logs covering the period of the incident and up to six months prior. The DOI CSO and Bureau or Office Incident Response Teams will determine when an incident requires additional focus and attention.

All Contractor employees and information system service providers shall report all information security incidents to the appropriate Bureau or Office Incident Response Team and to the DOI Computer Incident Response Center (DOI CIRC) immediately, no later than 1 hour after “becoming aware of the incident,” at DOICIRC@ios.doi.gov, (703) 648-5655, regardless of day or time. Any contractor action to investigate, identify, respond to, contain, remediate, or recover from a security event (which includes “suspected” incidents) constitutes “becoming aware of the incident.”

When notifying the DOI CIRC and the appropriate Bureau or Office Incident Response Team, the Contractor shall include the CO and COR on the correspondence email to DOICIRC@ios.doi.gov and the appropriate Bureau or Office Incident Response Team. If the Contractor reported the incident by phone, then immediately following the reporting, the Contractor shall notify the CO and COR via e-mail. The Contractor is responsible for verifying that all notification was received and acknowledged by the CO or COR. If the Contractor has any questions regarding these procedures, they shall contact the CO or the COR.

The Contractor shall not include any controlled information in the subject or body of any e-mail. Contractor shall transmit controlled information using validated encryption methods compliant with the current NIST FIPS 140 version to protect controlled information in e-mail attachments. Passwords must not be communicated in the same email as the attachment. Contractor shall contact the CO or COR if encryption software is needed for guidance on appropriate, federally approved encryption methods.

Section 3.0 – Privacy Requirements

A. Applicability.

This term applies to the Contractor, its subcontractors, and Contractor employees (hereafter referred to collectively as “Contractor”). The Contractor shall use these terms in all subcontracts.

B. Definitions.

As used in this term-

Breach: the loss of control, compromise, unauthorized disclosure, unauthorized acquisition, or any similar occurrence where (1) a person other than an authorized user accesses or potentially accesses PII or (2) an authorized user accesses or potentially accesses PII for an other-than-authorized purpose.

1. PII - information that can be used to distinguish or trace an individual's identity, either alone or when combined with other information that is linked or linkable to a specific individual.
2. Sensitive PII (SPII) is a subset of PII that, if lost, compromised, or disclosed without authorization could result in substantial harm, embarrassment, inconvenience, or unfairness to an individual. SPII is CUI and requires additional controls and safeguards to protect the privacy of individuals. Sensitivity of PII may depend on specific PII elements, groupings of PII, context or use of PII, or requirements under applicable federal laws, regulations, and policy such as the Privacy Act of 1974 and the CUI Rule. SPII includes but is not limited to full or truncated Social Security Number (SSN), driver's license or state identification number; financial account number; date of birth; citizenship; ethnic or religious affiliation; biometric identifiers such as fingerprint, voiceprint, or iris scan; medical information; criminal history; and system authentication information such as passwords, personal identification numbers or mother's maiden name.

C. Privacy Act Requirements

The Contractor shall comply with the requirements of the Privacy Act of 1974 and shall not remove PII or Privacy Act material from government facilities or systems, or facilities or systems operated or maintained on the government's behalf, without the express written permission of the Head of the Contracting Activity. When Privacy Act records, information, data, documentary material, and equipment is no longer required, the Contractor shall return it to the Department's control or hold it until otherwise directed. Items returned to the government shall be hand carried, mailed, emailed, or securely electronically transmitted to the CO.

D. Privacy Controls

The Contractor shall meet the privacy requirements of the E-Government Act, FISMA, applicable federal privacy laws, NIST, OMB, and DOI (including but not limited to those listed in the contract table of authorities) for the collection, handling, processing, and sharing of PII, and ensure compliance with the privacy controls requirements of NIST SP 800- 53, FIPS 199,

FIPS 200, and DOI Privacy Control standards as appropriate for the sensitivity of the system. The Contractor shall monitor the information system and update controls for continued privacy compliance on an ongoing basis and shall provide status reports to the Department upon request.

The Contractor shall be required to provide all necessary support to assist the Department in meeting the requirements of the Privacy Act and related laws and shall provide supporting documentation and access to information upon request by authorized agency officials. Support in this context includes timely responding to requests for information from the Department about the access, creation, collection, use, storage, maintenance, transmission, sharing, or disposition of PII on the Contractor's system or processes, and providing timely review of relevant compliance documents for factual accuracy.

E. Privacy Breach Reporting Requirements

The Contractor shall provide notice to DOI CIRC or the appropriate Bureau or Office Incident Response Team of any known or suspected breach of PII and meet breach reporting and mitigation requirements in accordance with the DOI Privacy Breach Response Plan or bureau or office applicable response plan. The report of a breach of PII shall not, by itself, be interpreted as evidence that the Contractor has failed to provide adequate information security safeguards for sensitive information or has otherwise failed to meet the requirements of the contract.

The Contractor shall report all known or suspected breaches in electronic or physical form to the CO, COR, and the DOI CIRC or bureau or office point of contact no later than one hour following discovery via email at DOICIRC@ios.doi.gov or (703) 648-5655, regardless of day or time and as specified within the DOI Privacy Breach Response Plan and the applicable bureau or office response plan.

The Contractor shall not include any controlled information in the subject or body of any e-mail and shall not include any SPII. Where necessary, SPII shall be transmitted using encryption methods to protect CUI in attachments in accordance with cryptographic requirements prescribed by the current NIST FIPS 140 version. Passwords shall be securely communicated separately.

The Contractor shall not contact individuals impacted by a breach involving PII until directed by the government. Department officials will make all determinations related to response activities, appropriate notifications to individuals and federal agencies and other organizations, and any remedial actions.

The Contractor shall take immediate action to contain and mitigate the impact of the breach and cooperate with DOI officials to investigate the breach and determine remedial measures as follows:

- The Contractor shall provide full access and cooperation for all activities—determined by the Department to be required—to ensure an effective breach response, including providing all requested images, log files, and event information to facilitate rapid resolution of breaches.
- Breach response activities—determined to be required by the Department—may include but are not limited to inspections, investigations, forensic reviews, data analyses and

processing, and final determinations of responsibility for the breach and/or liability for any additional response activities.

- The Department, at its sole discretion, may obtain the assistance of federal agencies and/or third-party firms to aid in breach response activities, as needed.

The Contractor shall provide a detailed written analysis with the following data elements available at the time of the breach, and a follow up report with any additional information or updates within 24 hours of the initial breach report, that shall address all relevant information concerning the compromise:

- Government programs, platforms, or systems involved
- Location(s) of the breach
- Date and time the breach was discovered
- Nature of the event (loss, theft, or unauthorized access)
- Description or summary of events
- Description of PII involved, such as name, Social Security number, date of birth, etc.
- Number of potentially impacted individuals
- Estimated number of records exposed or compromised

The Contractor shall collaborate with DOI officials on breach remedial activities to include notifications, credit monitoring, establishing a call center, or other relief to affected individuals as appropriate under the circumstances of the breach as prescribed in the contract. The Contractor shall have the capability to notify individuals whose PII resided in the Contractor system at the time of the breach not later than five business days after being directed by DOI officials to notify affected individuals. The Contractor shall not proceed with notifications unless the CO has directed in writing that notification is appropriate, subject to prior approval by DOI Privacy Officials, in accordance with the DOI Privacy Breach Response Plan. Notification may require the Contractor's use of address verification and/or address location services. All determinations, including response activities, notifications to affected individuals and/or federal agencies, and related services will be made by authorized DOI officials at DOI's discretion.

If an SPII or PII breach occurs because of the violation of a term of the contract by the Contractor or its employees, or the Contractor's covered persons, the Contractor shall, as directed by the CO and at no cost to the government, without delay correct or mitigate the violation. If the Department incurs a cost for individual notifications and remediation services procured as the result of a breach of SPII or PII, the COR will contact the CO to coordinate appropriate remedies per contract terms and conditions.

F. Privacy Training Requirements

All Contractor and subcontractor employees are required to take privacy awareness training prior to accessing DOI information or information systems. All Contractor and subcontractor employees must also complete role-based privacy training prior to accessing PII or Privacy Act data. All required privacy training shall be completed within 30 days of contract award and prior

to granting access to DOI information systems or any PII, and must be completed on an annual basis thereafter, not later than September 30th of each year. Any new Contractor employees assigned to the contract shall also meet these privacy training requirements. Privacy training shall meet the requirements of FAR 52.224-3 Privacy Training (January 2017).

The Contractor shall maintain copies of training certificates for all Contractor and subcontractor employees as a record of compliance. Initial training certificates for each Contractor and subcontractor employee shall be provided to the COR not later than 30 days after contract award. Subsequent training certificates to satisfy the annual training requirement shall be submitted to the COR via e-mail notification not later than October 30th of each year. The e-mail notification shall state the required training has been completed for all Contractor and subcontractor employees.

Section 4.0 – Accessibility Requirements (Section 508)

Section 508 of the Rehabilitation Act of 1973, as amended (29 U.S.C § 794 (d)) requires that when federal agencies develop, procure, maintain, or use information and communication technology (ICT), it must be accessible to people with disabilities. Federal employees and members of the public who have disabilities must have access to and use of information and data that is comparable to people without disabilities.

When ICT conforming to one or more requirements in the Revised 508 Standards is not commercially available, the agency shall procure the ICT that best meets the agency's business needs and the Revised 508 Standards, in accordance with 36 CFR 1194 E202.7 and FAR 39.2. Any selection of a product or service that meets less accessibility standards due to a significant difficulty or expense shall only be permitted under an undue burden claim and requires authorization from the DOI Section 508 Office according to 375 DM 8.

A. Applicability

This term applies to the Contractor, its subcontractors, and Contractor employees (hereafter referred to collectively as "Contractor"). The Contractor shall use these terms in all subcontracts.

B. Requirements

1. Products, platforms, and services delivered as part of a statement of work that are ICT, or contain ICT, shall conform to the Revised 508 Standards, which are located at 36 CFR § 1194.1 and Appendices A, C and D and available at <https://www.access-board.gov/ict/>. In the revised regulation, ICT replaced the term Electronic and Information Technology (EIT) used in the original 508 standards.
 - a. Applicable Functional Performance Criteria: All functional performance criteria in chapter 3 of the Revised 508 Standards apply when using an alternative design or technology that results in substantially equivalent or greater accessibility and usability by individuals with disabilities than would be provided by conformance to one or more of the requirements in chapters 4 and 5 of the Revised 508 Standards, or when chapters 4 or 5 do not address one or more functions of ICT.
 - b. Applicable 508 requirements for electronic content features and components (including electronic training materials): All requirements in E205 of the revised

508 Standards apply, including all Web Content Accessibility Guidelines (WCAG) 2.0 Level AA Success Criteria.

- c. Applicable 508 requirements for software features and components (including Software infrastructure): All requirements in chapter 5 apply, including all WCAG 2.0 Level AA Success Criteria, 502 Interoperability with Assistive Technology, and 503 Application.
 - d. Applicable 508 requirements for hardware features and components: All requirements in chapter 4 apply.
 - e. Applicable 508 requirements for support services and documentation: All requirements in chapter 6 apply.
2. When providing installation, configuration or integration services for ICT, the Contractor shall not reduce the original ICT item's level of Section 508 conformance prior to the services being performed.
 3. When providing maintenance upgrades, substitutions, and replacements to ICT, the Contractor shall not reduce the original ICT's level of Section 508 conformance prior to upgrade, substitution, or replacement. The agency reserves the right to request an Accessibility Conformance Report (ACR) for proposed substitutions and replacements prior to acceptance.
 4. Contractors shall provide an ACR for each commercially available ICT item offered through this contract. The ACR should be created using the Voluntary Product Accessibility Template (VPAT) Version 2.4 Rev 508 (or later), located at <https://www.itic.org/policy/accessibility/vpat>.
 5. When developing or modifying ICT for the government, the Contractor shall ensure the ICT fully conforms to the applicable Section 508 Standards. When modifying a commercially available or government-owned ICT, the Contractor shall not reduce the original ICT item's level of Section 508 conformance.
 6. When developing or modifying web and software ICT, the Contractor shall demonstrate Section 508 conformance by providing Section 508 test results based on the versions of the DHS Trusted Tester Methodology currently approved for use, as defined at <https://www.dhs.gov/compliance-test-processes>. The Contractor shall use testers who are certified by DHS on how to use the DHS Trusted Tester Methodology (e.g., "DHS Certified Trusted Testers") to conduct accessibility testing. Information on how testers can become certified is located at <https://www.dhs.gov/publication/trusted-tester-resources>.
 7. Contractor personnel shall possess the knowledge, skills, and abilities necessary to address the applicable revised Section 508 Standards for each ICT.
 8. Exceptions for statements of work have been determined by DOI and only the exceptions described in 375 Departmental Manual (DM) 8 DOI Section 508 Program and Responsibilities, dated February 5, 2016, may be applied. Any request for additional exceptions shall be sent to the COR and a determination will be made according to 375 DM 8.

Section 5.0 – Records Management Requirements

When federal agencies acquire goods or services, they need to determine what federal records management requirements should be included in the contract. Federal contractors often create, send, or receive federal records. Federal contracts should provide clear legal obligations describing how the contract employees must handle federal records.

Agency records officers, procurement counsel, and acquisitions officers must discuss how to integrate records management obligations into their existing procurement processes. The National Archives and Records Administration (NARA) has developed language to be included as an agency-specific term and condition in federal contracts for a variety of services and products. Most contracts should include language on records management obligations, but each contract should be evaluated individually as not all conditions will be appropriate for all contracts. For example, the data-rights paragraph may not be appropriate for all contracts. Instead, agencies may be better served by one of the established data-rights clauses in the Federal Acquisition Regulations.

This language should not replace specific records management requirements included within federal information system contracts.

A. Applicability

This term applies to all Contractors whose employees create, work with, or otherwise handle federal records, as defined in section B, regardless of the medium in which the record exists. The Contractor shall use these terms in all subcontracts.

B. Definitions

Federal record – (as defined in 44 U.S.C. § 3301) includes all recorded information, regardless of form or characteristics, made or received by a federal agency under federal law or in connection with the transaction of public business and preserved or appropriate for preservation by that agency or its legitimate successor as evidence of the organization, functions, policies, decisions, procedures, operations, or other activities of the United States Government or because of the informational value of data in them.

The term federal record:

- includes all departmental records
- does not include personal materials
- applies to records created, received, or maintained by Contractors pursuant to their Department contract
- may include deliverables and documentation associated with deliverables

C. Requirements

1. Contractors shall comply with all applicable records management laws and regulations, as well as NARA records policies, including but not limited to the Federal Records Act (44 U.S.C. chapters 21, 29, 31, 33), NARA regulations at 36 CFR Chapter XII Subchapter B, and Departmental policy.

2. The Contractor shall coordinate with the appropriate Departmental and Bureau Records Management Programs to ensure the identification, storage, retrieval, preservation, access, control markings, and disposition of records using the official Departmental record keeping system(s). This includes the implementation of all technical aspects to appropriately interconnect and or transfer records to the system(s). Any exceptions to using this system must be expressly presented by the Contractor in writing to and approved by the Departmental and/or Bureau Records Management Programs. Thereafter, each major or minor variance must be expressly presented in writing.
3. When the Department or a bureau contracts for the design, development, or operation of a system on behalf of the Department, the Contractor must make and preserve records to sufficiently document all planning, design, development, or operation of the system.
4. In accordance with 36 CFR 1222.32, all data created for government use and delivered to, or falling under the legal control of, the government are federal records subject to the provisions of 44 U.S.C. chapters 21, 29, 31, and 33, the Freedom of Information Act (FOIA) (5 U.S.C. 552), as amended, the Privacy Act of 1974 (5 U.S.C. 552a), as amended, and Executive Order 13556 Controlled Unclassified Information. Contractors shall maintain all records created for government use or created while performing the contract and/or delivered to, or under the legal control of the government and must manage them in accordance with federal law. Electronic records and associated metadata must be accompanied by sufficient technical documentation to permit understanding and use of the records and data and must be managed and scheduled for disposition only as permitted by statute or regulation.
5. The Department and its Contractors are responsible for preventing the alienation or unauthorized destruction of records, including all forms of mutilation. Records may not be removed from the legal custody of DOI or destroyed except for in accordance with the provisions of the agency records schedules and with the written concurrence of the Head of the Contracting Activity and the Bureau Records Officer. The routine disposition of records created and maintained by DOI and the Contractor may be suspended through coordination of the Departmental Records Officer, Bureau or Office Records Officer, Records Management Contact, Program Manager, or Site Manager. Willful and unlawful destruction, damage or alienation of federal records is subject to the fines and penalties imposed by 18 U.S.C. 2701. In the event of any unlawful or accidental removal, defacing, alteration, or destruction of records, the Contractor must report to DOI. The agency must report promptly to NARA in accordance with 36 CFR 1230.
6. The Contractor shall immediately notify the appropriate CO upon discovery of any inadvertent or unauthorized disclosures of information, data, documentary materials, records, or equipment. Disclosure of non-public information is limited to authorized personnel with a need-to-know as described in the contract vehicle. The Contractor shall ensure that the appropriate personnel, administrative, technical, and physical safeguards are established to ensure the security and confidentiality of this information, data, documentary material, records and/or equipment is properly protected. The Contractor shall not remove material from government facilities or

- systems, or facilities or systems operated or maintained on the government's behalf, without the express written permission of the Head of the Contracting Activity which the Contractor requests from the CO. When information, data, documentary material, records and/or equipment is no longer required, the Contractor shall return it to DOI control or must hold it until otherwise directed. Items returned to the government shall be hand carried, mailed, emailed, or securely electronically transmitted to the CO or address prescribed in the contract vehicle. The Contractor shall provide a mechanism for reliably deleting DOI data upon request by the Department.
7. The Contractor is required to obtain the CO's approval prior to engaging in any contractual relationship (sub-contractor) in support of this contract requiring the disclosure of information, documentary material and/or records generated under, or relating to, contracts. The Contractor (and any sub-contractor) is required to abide by government and DOI guidance for protecting sensitive, proprietary information, classified, and CUI.
 8. The Contractor shall only use government IT equipment for purposes specifically tied to or authorized by the contract and in accordance with DOI policy.
 9. The Contractor shall not create or maintain any records containing any non-public DOI information that are not specifically tied to or authorized by the contract.
 10. The Contractor shall not retain, use, sell, or disseminate copies of any deliverable that contains information covered by the Privacy Act of 1974 or that which is generally protected from public disclosure by an exemption to the Freedom of Information Act.
 11. The Department owns the rights to all data and records produced as part of this contract. All deliverables under the contract are the property of the U.S. Government for which DOI shall have unlimited rights to use, dispose of, or disclose such data contained therein as it determines to be in the public interest. Any Contractor rights in the data or deliverables must be identified as required by FAR 52.227-11 through FAR 52.227-20.
 12. Training. All Contractor employees assigned to this contract who create, work with, or otherwise handle records are required to take DOI-provided records management training. The Contractor is responsible for confirming training has been completed according to agency policies, including initial training and any annual or refresher training.

Section 6.0 – Contractor Personnel Security Requirements

Acquired services shall comply with the following regulations and requirements. Homeland Security Presidential Directive-12 requires that all federal entities ensure that all Contractors gaining physical access to federally controlled facilities or logical access to federally controlled information systems have current and approved security background investigations that are equivalent to investigations performed on federal employees; see section 1.0, part C of this document. The Contractor shall comply with the contract terms and conditions. The Office of Personnel Management (OPM) will perform background investigations.

Section 7.0 – Controlled Unclassified Information Requirements

A. Applicability

This term applies to the Contractor, its subcontractors, and Contractor employees (hereafter referred to collectively as “Contractor”). The Contractor shall use these terms in all subcontracts.

B. Definitions

1. Sensitive Information - any information that warrants a degree of protection and administrative control as defined in law, regulation, governmentwide policy, or judicial opinion.
2. Controlled Unclassified Information (CUI) - information the government creates or possesses, or that an entity creates or possesses for or on behalf of the government, that a law regulation or governmentwide policy requires or permits an agency to handle using safeguarding or dissemination controls. Legacy materials may retain For Official Use Only (FOUO), Sensitive But Unclassified (SBU), or other markings until the artifact is no longer considered “data at rest;” it is revised, reused, or goes in motion.
3. Personally Identifiable Information (PII) - a subset of information designated as CUI. See section 3.0, Privacy Requirements, for additional requirements.
4. Sensitive PII - subset of PII that requires additional controls and safeguards. See section 3.0, Privacy Requirements, for additional requirements.

C. Authorities

1. EO 13556, Controlled Unclassified Information, November 4, 2010
2. 32 CFR 2002, Controlled Unclassified Information (Implementing Directive)
3. The Federal CUI Registry
4. Departmental standards

D. Requirements

1. The Contractor must manage CUI in accordance with applicable laws, regulations, executive orders, and policies.
2. The Contractor must identify, and safeguard controlled information from unauthorized release into public domain, or to unauthorized persons, organizations, or subcontractors.
3. The Contractor or subcontractor must not disclose or release the materials provided to the Contractor to any individuals of the Contractor's organization not directly engaged in providing services under the contract or that do not have a valid need-to-know. All technical data provided to the Contractor by the government must be protected from public or private disclosure in accordance with the markings printed on them.
4. The Contractor must not release to anyone outside the Contractor’s organization any sensitive information unless the CO has given prior written approval. This includes, but is not limited to, news releases, marketing promotions, articles, interviews,

reports, social media posts, and any other media releases. Requests for approval must identify the specific information to be released, the medium to be used, the purpose for the release, and a description of the need-to-know. The Contractor must submit its request to the CO at minimum ten business days before the proposed date for release. Subcontractors must submit requests for authorization to release through the prime Contractor to the CO.

5. Contractor controlled computer systems housing CUI must follow the NIST SP 800-171 Rev. 2 (or current Rev); SP 800-171A; and SP 800-172 (or current Rev).
6. If the Contractor believes that CUI information has been compromised, they will follow the incident response procedures in the DOI Incident Response Policy and Plan.

E. Training

Individuals originating or working with sensitive information must complete CUI training to become an authorized holder. CUI must only be disseminated to authorized holders in the furtherance of a lawful government purpose.

Section 8.0 – Paperwork Reduction Act of 1995

If a portion of the contract seeks information from ten or more members of the public, the contractor will work with the Bureau Information Collection Clearance Officer to obtain a Paperwork Reduction Act clearance from the OMB.

Section 9.0 – Reference Documents

All DOI-specific documents referenced in this document necessary for Contractor review and reference can be provided by the CO/COR.

Section 10.0 – Glossary

Please refer to the NIST online glossary at: <https://csrc.nist.gov/glossary>.